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Natural xanthonenes as α -Mangostin induce vasorelaxation involving key gating residues in the S6 domain of BK channels

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eLife Assessment

The present manuscript by Cordeiro et al., shows **convincing** evidence that α -mangostin, a xanthone obtained from the fruit of the *Garcinia mangostana* tree, behaves as a strong activator of the large-conductance (BK) potassium channels. The authors suggest that α -mangostin activation of the BK channel is state-independent, and molecular docking and mutagenesis suggest that α -mangostin binds to a site in the internal cavity. Additionally, the authors show that α -mangostin can relax arteries, further suggesting the plausibility of the proposed effects of this compound. These are **valuable** findings that should be of interest to channel biophysicists and physiologists alike.

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Abstract

Polyphenolic compounds are widely explored for health benefits, including hypertension, but their active ingredients, molecular targets, and mechanisms remain poorly defined. We identify the xanthone Mangostin from *Garcinia mangostana* as a potent modulator of several potassium channels, with large-conductance K^+ (BK) channels as its primary target for vasorelaxation. Mangostin activated BK channels as α subunits alone, in complexes with vascular β_1 subunits, and in reconstituted $BK\alpha/\beta_1$ - Ca_v nanodomains. It shifted BK voltage activation to more negative potentials by antagonizing channel closure and promoting channel opening without markedly altering Ca^{2+} sensitivity. Docking, competition, single channel analysis and mutagenesis localized the binding site in the pore cavity below the SF, involving gating-critical S6 residues I308, L312, and A316, and suggest that Mangostin stays bound in closed and open states. These findings establish BK channel activation as the core molecular mechanism driving Mangostin's vascular effects and define its structural mode of action, informing nutraceutical safety assessment and BK-targeted drug design.

Introduction

Nutraceuticals (superfoods, dietary supplements, and traditional remedies) promise a plethora of health benefits that are usually not definitely proven by clinical studies. Well-known examples are resveratrol present in grapes and red wine (Jang *et al.*, 1997 [DOI](#); Brown *et al.*, 2024 [DOI](#)), carotenes and tocopherols (Alpha-Tocopherol, Beta Carotene Cancer Prevention Study Group, 1994 [DOI](#); Hennekens *et al.*, 1996 [DOI](#); Xin *et al.*, 2022 [DOI](#)), or the recent resurgence of quercetin (Lee *et al.*, 2021 [DOI](#); Di Pierro *et al.*, 2023 [DOI](#)). In evaluating clinical relevance and explore pharmacological opportunities of such phytochemicals, obstacles are often the identification of the biologically active substance in complex matrices of plant secondary metabolites and the lack of a definite

cellular determinant that could link a biologically active substance to a physiological process. However, a clear molecular target and mechanism of action that can explain its effects is the prerequisite to further drug development.

Natural polyphenols are known to modulate diverse potassium (K^+) channel families. For example, various members of the flavonoid subgroup activate K_{Ca} (BK) channels as well as several K_v , K_{IR} , and K_{2P} channels, including TREK-1 channels (Gierten *et al.*, 2008; Nardi and Olesen, 2008; Kim, Kang and Han, 2011; for a comprehensive review see Richter-Laskowska *et al.*, 2023). A recent study suggested that α -Mangostin, a xanthone from *Garcinia mangostana*, modulates ion channels and binds in the pore cavity of TREK K_{2P} channels (Kim *et al.*, 2023). α - and γ -Mangostins are the main xanthones found in the mangostane fruit pericarp, whose ethanolic extracts are used as traditional medicine and consumed as nutraceuticals. They are thought to exert a plethora of health-promoting effects, such as cardioprotective (Eisvand *et al.*, 2022), analgesic (Kim *et al.*, 2023), antioxidant (Kong, Jia and Jia, 2022), antiinflammatory (Kim *et al.*, 2021), antidiabetic (John *et al.*, 2022), antifibrotic (Li *et al.*, 2019), antimicrobial (Tatiya-Aphiradee, Chatuphonprasert and Jarukamjorn, 2016), and antiproliferative effects (Majdalawieh *et al.*, 2024). Interestingly, very recent research reported that α -Mangostin has antihypertensive properties, and it decreased systolic and diastolic blood pressure in a rat model (Xu *et al.*, 2024). Earlier studies had demonstrated the relaxation of aortic rings upon Mangostin incubation, but remained somewhat contradictory as to the site of action and the molecular target (Chairungsrilerd *et al.*, 1996, 1997; Tep-Areenan and Suksamrarn, 2012). Hence, the molecular mechanism of this antihypertensive effect of Mangostin remains unclear.

Major potassium channels present in vascular smooth muscle are Ca^{2+} -activated BK channels, also known as MaxiK, Slo1 or $K_{Ca}1.1$ channels (Wu and Marx, 2010; Tykocki, Boerman and Jackson, 2017; Pereira da Silva *et al.*, 2022). They are synergistically activated by intracellular calcium (Ca^{2+}) and voltage, and they play a crucial role in the regulation of vascular smooth muscle tone, with implications for blood pressure regulation. BK channels act as feedback regulators which counteract depolarization and balance the elevation of Ca^{2+} through voltage-dependent L-type Ca^{2+} channels. The pore-forming BK α subunit is expressed ubiquitously, and tissue specific properties are conveyed by regulatory β -, γ -, and LINGO subunits (Latorre *et al.*, 2017; Gonzalez-Perez and Lingle, 2019; Dudem *et al.*, 2020). In vascular smooth muscle cells, BK α assembles with the β 1 subunit, which enhances its apparent Ca^{2+} sensitivity, decreases voltage sensitivity (Dworetzky *et al.*, 1996; Brenner *et al.*, 2000; Orio and Latorre, 2005; Li and Yan, 2016), and modulates its pharmacological properties (McManus *et al.*, 1995; Hanner *et al.*, 1997).

Structurally, BK channels are homotetramers consisting of the pore-forming α subunits with 7 transmembrane segments (S0-S6), a linker region, and two intracellular RCK domains. The RCK domains and their interface to the transmembrane segments contain different Ca^{2+} binding sites, S1-S4 comprise the voltage sensing domain, and S5 and S6 together with the pore helix constitute the pore domain. Both depolarization or Ca^{2+} elevation can open the pore via allosteric mechanisms, and this regulation as well as the modulation by mutations, small molecules, and regulatory subunits can be described well by an allosteric gating model (Horrigan and Aldrich, 2002; Latorre *et al.*, 2017). However, the structural determinants that underlie this allosteric modulation remain unclear. The S6 segment has been shown to undergo conformational changes during activation, and several gating sensitive residues as well as modulator binding sites for molecules of surprisingly different chemical structure are known in this pore region or the adjacent linker to the cytoplasmic C-terminus (Nardi and Olesen, 2008; Zhou, Xia and Lingle, 2011; Roy *et al.*, 2012; Hoshi, Xu, *et al.*, 2013; Chen, Yan and Aldrich, 2014; Webb *et al.*, 2015; Hoshi and Heinemann, 2016; Schewe *et al.*, 2019; Rockman, Vouga and Rothberg, 2020; Gonzalez-Sanabria *et al.*, 2025). Investigations of the activation mechanism indicated that these substances act without directly affecting the voltage- or Ca^{2+} sensory domains, probably by modifying the close interactions of S5 and S6 with the voltage sensor bundle and the gating ring that enable opening of the channel.

Small molecule BK channel activators are promising to treat diverse diseases like fragile X syndrome, overactive bladder, pulmonary hypertension, chronic obstruction, erectile dysfunction, and reperfusion injury; however, no prospective drug successfully completed a phase III trial yet (Bentzen *et al.*, 2014 [DOI](#); Barenco-Marins *et al.*, 2023 [DOI](#); Ferraguto *et al.*, 2024 [DOI](#)). Therefore, adding a pharmacophore derived from the xanthone group may aid in developing future drug candidates.

We used a combination of functional studies in cultured cells and vascular tissue as well as molecular docking to show that BK channels are a molecular target of Mangostins and that they mediate vascular relaxation observed upon Mangostin application. BK channels were opened most prominently among the K⁺ channel representatives tested and were activated by α-Mangostin as well as by an extract of the mangosteen pericarp marketed as a nutraceutical. Investigation of the activation mechanism showed that α-Mangostin essentially facilitates BK channel activation by shifting its voltage dependence, modulating gating kinetics, and enhancing the open probability, without changing the Ca²⁺ sensitivity. These effects were due to direct binding to the pore-forming BKα subunit. Competition experiments, molecular docking, and scanning mutagenesis identified the binding region in S6 just below the selectivity filter (SF). Mangostin activation was preserved in reconstructed nanodomains with Ca_v channels, which cause local elevation of intracellular Ca²⁺, mimicking the physiological situation in smooth muscle cells. Finally, we show that BK channels specifically mediate relaxation of aortic preparations from mice. Our study explains the molecular activation of BK channels by a natural xanthone and sheds light on the mechanism underlying one of its claimed health benefits.

Results

α-Mangostin particularly activates BKα and BKα/β1 channels

In previous studies, α-Mangostin was shown to activate members of the K_{2P} channel family, while other K⁺ channels were inhibited (Kim *et al.*, 2021 [DOI](#), 2023 [DOI](#)). We aimed to obtain a broader pharmacological profile of α-Mangostin in K⁺ channels and we investigated members of all six subfamilies of K_{2P} channels as well as representatives of the K_v and K_{ir} channel families. Confirming the previous studies, we observed a strong activation of TREK-1 K_{2P} channels (fold change 10.33 ± 1.64; Fig. 1A [DOI](#) and Fig. S1 [DOI](#)) and inhibition of TRESK K_{2P} channels as well as K_v1.3 channels by ≈50% and ≈60 %, respectively (Fig. 1A, B [DOI](#) and Fig. S1 [DOI](#)). Furthermore, we identified three additional targets of α-Mangostin: strongly inhibited were the K_{2P} members TWIK-1^m (≈88 %), TWIK-2^m (≈50 %), and TASK-3 (≈80 %) (Fig. 1A, B [DOI](#) and Fig. S1 [DOI](#)). In contrast, BK channels were most potently activated, with a fold change of 20.41 ± 2.39 for BKα channels and of 24.42 ± 3.85 for BKα/β1 channels, which is the complex predominantly present in smooth muscle cells of the vasculature (Fig. 1A, C [DOI](#)). This suggested that BK channels, especially the BKα/β1 heteromer, could be the molecular target mediating the vasorelaxant Mangostin effects reported in above studies.

The preparations from *Garcinia mangostana* plants that are consumed as ‘nutraceuticals’ contain an unstandardized mixture of secondary plant metabolites, among them other Mangostin derivatives. Therefore, we also investigated their modulatory effect on TREK-1, BKα and BKα/β1 currents (Fig. 1D [DOI](#)). The very similar γ-Mangostin activated BKα channels with slightly higher efficacy (fold change of 38.54 ± 4.8) than TREK-1 and BKα/β1 channels (fold change of 11.35 ± 1 and 29.32 ± 5.03). Interestingly, the dietary supplement (solubilized equivalent to 10 μM α-Mangostin) activated TREK1, BKα and BKα/β1 channels comparably to pure α-/γ-Mangostin (fold change of 9.18 ± 2.74, 19.98 ± 2.99 and 20.92 ± 3.49; Fig. 1D [DOI](#)), highlighting that mangostane xanthenes retain their activity in such extracts.

The application of different α-Mangostin concentrations resulted in a fast and dose-dependent activation of BKα and BKα/β1 channels, with a similar apparent EC₅₀ of 3.16 ± 1.13 μM and 2.57 ± 1.32 μM, suggesting that the potency of α-Mangostin is not affected by the presence of the β1-subunit (Fig. 1E [DOI](#)). However, we frequently noticed a decline of currents at concentrations >12 μM and limited solubility of higher Mangostin concentrations, which may imply that the actual maximal activation could not be reached.

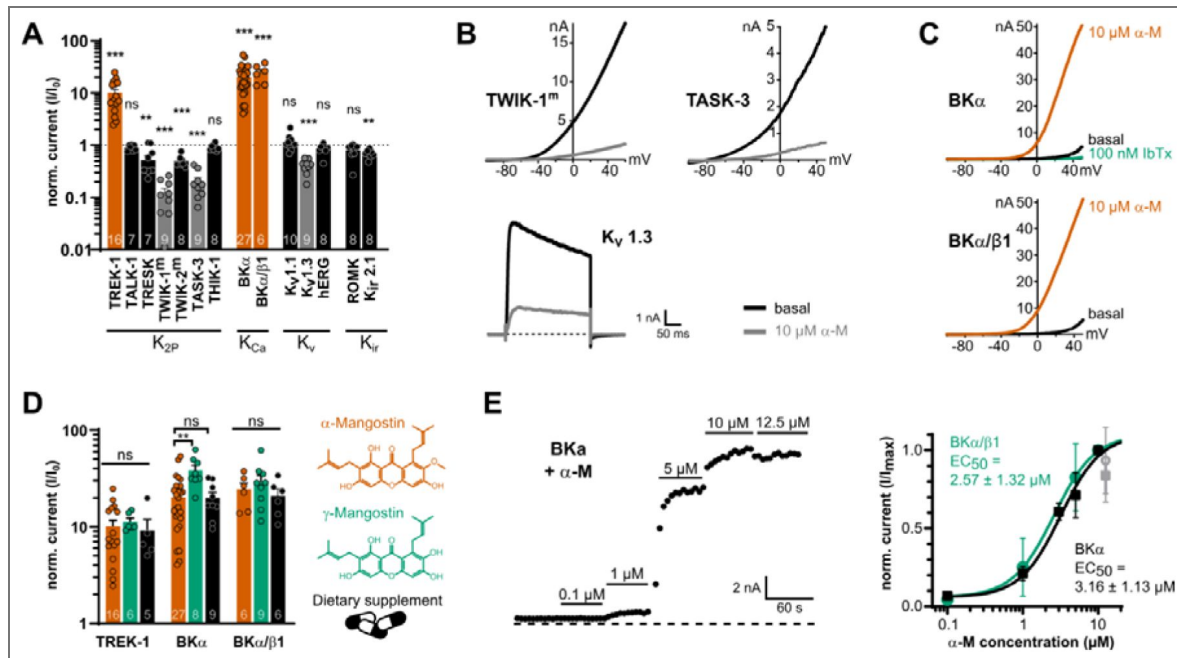


Figure 1. Mangostin potentially activates BK α and BK α / β 1 channels compared to other potassium channel representatives.

(A) Current fold change $\pm$ SEM of currents of representatives from different potassium channel families upon application of 10 μ M α -Mangostin. K_{2P} and K_{Ca} channel currents were recorded with ramp protocols from -100 mV to +50 mV and analyzed at +40 mV; K_V1.1 and K_V1.3 channel currents were evoked using a rectangle pulse to +40 mV, and hERG currents were recorded with a rectangle pulse to +60 mV followed by hyperpolarization to -120 mV, which was analyzed. Kir channels were recorded using ramp protocols from -150 mV to +50 mV, and currents were analyzed at -140 mV. All measurements were made in transiently transfected HEK293 cells in physiological potassium gradients with 100 nM intracellular free Ca²⁺ with a holding potential of -80 mV. TWIK-1^m and TWIK-2^m denote channels where the retrieval motif was removed to improve membrane expression, and intracellular K⁺ was exchanged for Rb⁺ to enhance currents. **(B)** Representative current traces of channels that were inhibited more than 60 % by 10 μ M α -Mangostin. **(C)** Representative current traces of BK α and BK α / β 1 channels activated by 10 μ M α -Mangostin (α -M). **(D)** Current fold change $\pm$ SEM after application of α -Mangostin, γ -Mangostin, and a dietary supplement to TREK-1, BK α and BK α / β 1 channels. Currents were recorded and analyzed as in (A). **(E)** Representative time course of the dose-dependent activation of BK α channels by increasing concentrations of α -Mangostin (left) and the resulting dose-response-relationships for BK α and BK α / β 1 channels (right). Currents were recorded and analyzed as in (A); the grey data point at 12.5 μ M was not included in the Hill fit. Data and statistics see [Tables S1](#) and [S2](#).

α -Mangostin shifts the voltage activation of BK α and BK α / β 1 channels to more negative values

Having established the prominent activation of BK α and BK α / β 1 channels by α -Mangostin, we wanted to gain more insight into its activation mechanism. The open probability of BK channels is controlled by voltage signals, calcium signals, or shift of the closed-open equilibrium (e.g. by mutations or modulators), and the current further depends on the single channel amplitude and the time the channel adopts an open conductive state. We therefore investigated these aspects of BK channel gating in macroscopic and single-channel measurements.

As we already observed in the ramp measurements, the threshold of BK voltage activation was shifted to more negative potentials upon α -Mangostin application (Fig. 1C). Therefore, we first examined if α -Mangostin affects the voltage activation of BK channels. We used a step protocol over a range of potentials followed by a repolarization step to elicit tail currents in symmetrical bi-ionic conditions where intracellular K⁺ was replaced with Cs⁺ to reduce the strong outward currents in whole-cell experiments (Fig. 2A; Cs⁺ does not alter voltage activation (Piskorowski and Aldrich, 2006)). We quantified the change in voltage activation induced by α -Mangostin by calculating the voltage of half-maximal activation ($V_{1/2}$) from the Boltzmann fit of conductance-voltage (GV) relationships (Fig. 2B). The application of 10 μ M α -Mangostin caused a substantial shift of the $V_{1/2}$ by 53.08 ± 4.9 mV to more negative voltages in BK α channels (from 110.45 ± 2.69 mV to 57.37 ± 3.6 mV). This effect was even more pronounced in smooth muscle-like BK α / β 1 channels, where $V_{1/2}$ shifted by 82.42 ± 4.96 mV to more hyperpolarized voltages (from 147.25 ± 5.66 mV to 64.83 ± 4.25 mV). The slope of the Boltzmann fit was not different before and after α -Mangostin activation in both channels (Fig. 2B, insets). Hence, BK α and BK α / β 1 channels already opened at less depolarized voltages in the presence of α -Mangostin, whereas voltage activation itself was not affected.

α -Mangostin predominantly affects deactivation of BK α and BK α / β 1 channels

An inspection of the currents elicited by the families of rectangle pulses shows that gating kinetics are altered in the α -Mangostin activated state (Fig. 2A). The shift in $V_{1/2}$ towards more negative potentials upon α -Mangostin activation could be caused either by an acceleration of channel activation, or by a slowing down of channel deactivation, or a combination of both. We therefore analyzed the kinetics of activation and deactivation by fitting monoexponential functions to the time course of outward and tail currents to obtain their time constants (τ). BK α channels are characterized by fast activating current and a fast deactivation visible as decay of the tail current, while the presence of the β 1 subunit slows these kinetics down (Fig. 2A, bottom; Dworetzky *et al.*, 1996). After application of 10 μ M α -Mangostin, activation of BK α channels was moderately accelerated 1.7-fold at +100 mV, with a τ of activation of 7.96 ± 1.64 ms before and 4.71 ± 0.71 ms afterwards (Fig. 2C). Additional measurements in 10 μ M Ca²⁺ where outward currents are present at lower voltages showed a bigger effect with less depolarization (0-100 mV; Fig. S2). The τ for deactivation at -50 mV increased $\approx$ 7-fold from 0.9 ± 0.04 ms to 6.85 ± 1.11 ms at +100 mV (Fig. 2C; S2). Again, α -Mangostin action was more pronounced in the smooth muscle-like BK α / β 1 channels, where activation was more than 5-fold faster with τ values of 63.76 ± 16.03 ms before and 12.36 ± 1.20 ms after α -Mangostin application. Deactivation kinetics were most affected with a substantial $\approx$ 27-fold increase in τ from 3.60 ± 0.16 ms to 95.6 ± 13.99 ms (Fig. 2C). These results suggest that the mechanism of α -Mangostin action differentially affects gating transitions that are associated with activation and deactivation.

As BK channels can be massively activated by Ca_i²⁺ under physiological conditions, we further obtained GV-relationships in the basal and in the activated state for higher Ca_i²⁺ concentrations relevant in smooth muscle cells to test if α -Mangostin activation is independent of Ca_i²⁺ concentration (Fig. 2D). α -Mangostin shifted the $V_{1/2}$ in 10 μ M free Ca_i²⁺ by 50.55 ± 9.13 mV to more negative potentials, comparable to the resting state with 0.1 μ M free Ca_i²⁺. However, in 1 μ M Ca_i²⁺ α -Mangostin the V shift was slightly higher, resulting in a shift by 82.74 ± 10.17 mV. This is

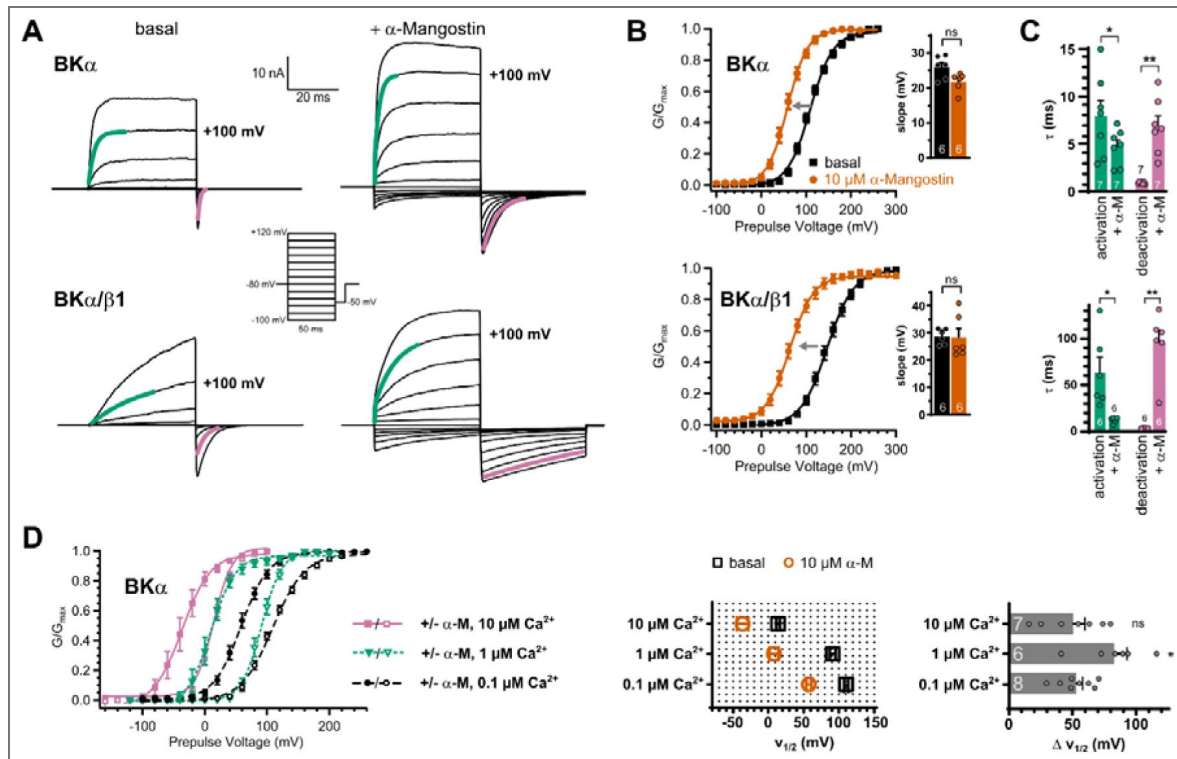


Figure 2. Effects of α -Mangostin on $BK\alpha$ and $BK\alpha/\beta1$ channel gating.

(A) Representative current traces for $BK\alpha$ and $BK\alpha/\beta1$ channels before and after activation by $10 \mu M$ α -Mangostin. Cells were measured in symmetrical 140 mM bi-ionic conditions with 140 mM Cs^+ as the intracellular ion and 100 nM free Ca^{2+} . Currents were elicited by a family of rectangle pulses from -100 mV to up to +300 mV in 20 mV increments from a holding potential of -80 mV, followed by repolarization to -50 mV to elicit inward tail currents. (B) GV-relationships for $BK\alpha$ and $BK\alpha/\beta1$ channels in the basal state and activated by $10 \mu M$ α -Mangostin, derived from tail current analysis of recordings as in (A). The grey arrow illustrates the shift of voltage activation towards more negative voltages caused by α -Mangostin and the inset shows the slope $\pm$ SEM of the Boltzmann fits. (C) Activation and deactivation kinetics of $BK\alpha$ and $BK\alpha/\beta1$ channels in the basal state and after activation by $10 \mu M$ α -Mangostin. The $\tau \pm$ SEM of activation/deactivation was determined from exponential fits to the current traces at +100 mV, as shown by the green and purple colored lines in panel (A). (D) GV-relationships of $BK\alpha$ channels in different free Ca^{2+} concentrations before and after activation by $10 \mu M$ α -Mangostin, derived from tail current analysis as above. Pulse voltages were -100 mV to +300 mV/ repolarization to -50 mV from a holding potential of -80 mV for 100 nM free Ca^{2+} , -120 mV to +200 mV/ repolarization to -50 mV from a holding potential of -80 mV for 1 μM free Ca^{2+} , and -160 mV to +100 mV/ repolarization to -80 mV from a holding potential of -120 mV for 10 μM free Ca^{2+} , in symmetrical 140 mM bi-ionic conditions with 140 mM Cs^+ as the intracellular ion. The $V_{1/2}$ values $\pm$ SEM before and after α -Mangostin application and the resulting shifts ($\Delta V_{1/2} \pm$ SEM) are shown in the bar graphs. Data and statistics see Tables S3 – S6.

consistent with previous studies, as BK α channel gating is primarily voltage-driven at very low Ca_i^{2+} concentrations and Ca_i^{2+} driven at high concentrations, while in the low-micromolar range (i.e., around 1 μM), Ca^{2+} binding sites are partially occupied, resulting in nonlinear gating where enhanced modulator effects can occur (Cui, Cox and Aldrich, 1997; Magleby, 2003; Clay, 2017).

α -Mangostin activation mechanism on single-channel and macroscopic current level

The marked slowing of deactivation visible as the slow tail current decay in macroscopic current recordings suggests that individual channels are open for longer periods of time in the presence of α -Mangostin. To investigate the effect of α -Mangostin on individual channels, we recorded single channel currents in excised inside-out patches from HEK293 cells in the presence of 100 nM free Ca_i^{2+} (Fig. 3A). Under these conditions, the open probability (P_o) of BK channels at a potential of +40 mV was very low (0.002 ± 0.0008), but rose to 0.77 ± 0.08 after activation by 10 μM α -Mangostin (Fig. 3B) with only very brief closings (Fig. 3A inset). The single channel amplitude derived from the all-points-histogram was not different between basal and α -Mangostin activated states (9.18 ± 0.29 pA and 9.76 ± 0.26 pA; Fig. 3B). We constructed dwell time distributions to determine the differences in the duration that a single channel resides in the closed or open state (Fig. 3C). The Log closed dwell time distributions showed a large shift of the main component to shorter dwell times, from 2.38 ± 0.23 (237 ms) to -1.39 ± 0.36 (less than 0.1 ms). The open dwell time distribution revealed that the log open time was markedly reduced from -0.84 ± 0.45 to 0.8 ± 0.24 (i.e. from 0.14 ms to 6.28 ms).

BK channels are known to show bursting gating behavior (Geng and Magleby, 2015). As bursts did not occur in our measurements in resting conditions, we elevated free Ca_i^{2+} to be able to perform burst analysis (Fig. 3D). Mean burst duration was increased and the long closures between bursts were shortened, explaining the increase in overall open probability (Fig. 3E). The intraburst kinetic distributions showed that α -Mangostin increased intraburst closed time from 20.8 ± 1.2 ms to 29.4 ± 1.3 ms, thereby reducing very brief closures (flicker), indicating that transitions became slower (Fig. 3F). The higher open probability within bursts resulted from both prolonged opening (as open dwell times increased from 18.6 ± 1.2 ms to 28.9 ± 1.2 ms) and from an increased probability of reopening (visible as increased number of openings per burst; Fig. 3E), consistent with an open-state stabilization that explains the reduced deactivation seen in macroscopic recordings.

The fact that the open and closed times were both affected in the overall dwell time distributions as well as in intraburst distributions suggests that α -Mangostin can bind to the open and the closed state. Accordingly, application of α -Mangostin on closed BK-channels (at -80 mV) resulted in maximal activation with the first depolarization pulse, indicating that α -Mangostin reached its site of action during the closed period (Fig. 3G).

Localization of the α -Mangostin binding site

We recently described the polypharmacology of the class of negatively charged activators (NCA) in different potassium channels, specifically in TREK-1 and BK channels. The BK channel opener GoSlo-SR-5-6 (Roy *et al.*, 2012; Webb *et al.*, 2015) also activated TREK-1 channels, while BL-1249 known as TREK/TRAAK channel opener, also activated BK channels. We identified their common binding site in the pore close to the fenestration of TREK-1 channels and Molecular Dynamics (MD) simulations predicted the equivalent binding site in the pore of BK channels (Schewe *et al.*, 2019). The question arose if α -Mangostin could also occupy this binding site, as in part suggested in a docking by (Kim *et al.*, 2023), who proposed that P183 and L304, which both are part of the NCA binding site, interact with α -Mangostin in TREK-1 channels. Therefore, we measured the dose-dependent α -Mangostin activation of TREK-1 wildtype channels and its 'signature' mutant of the fenestration, L304C. As expected, the apparent affinity is markedly reduced in the L304C mutant in the measurable concentration range (Fig. S3A). Quaternary ammonium ions are known to occupy the central cavity of K^+ channels as TREK-1 and BK just

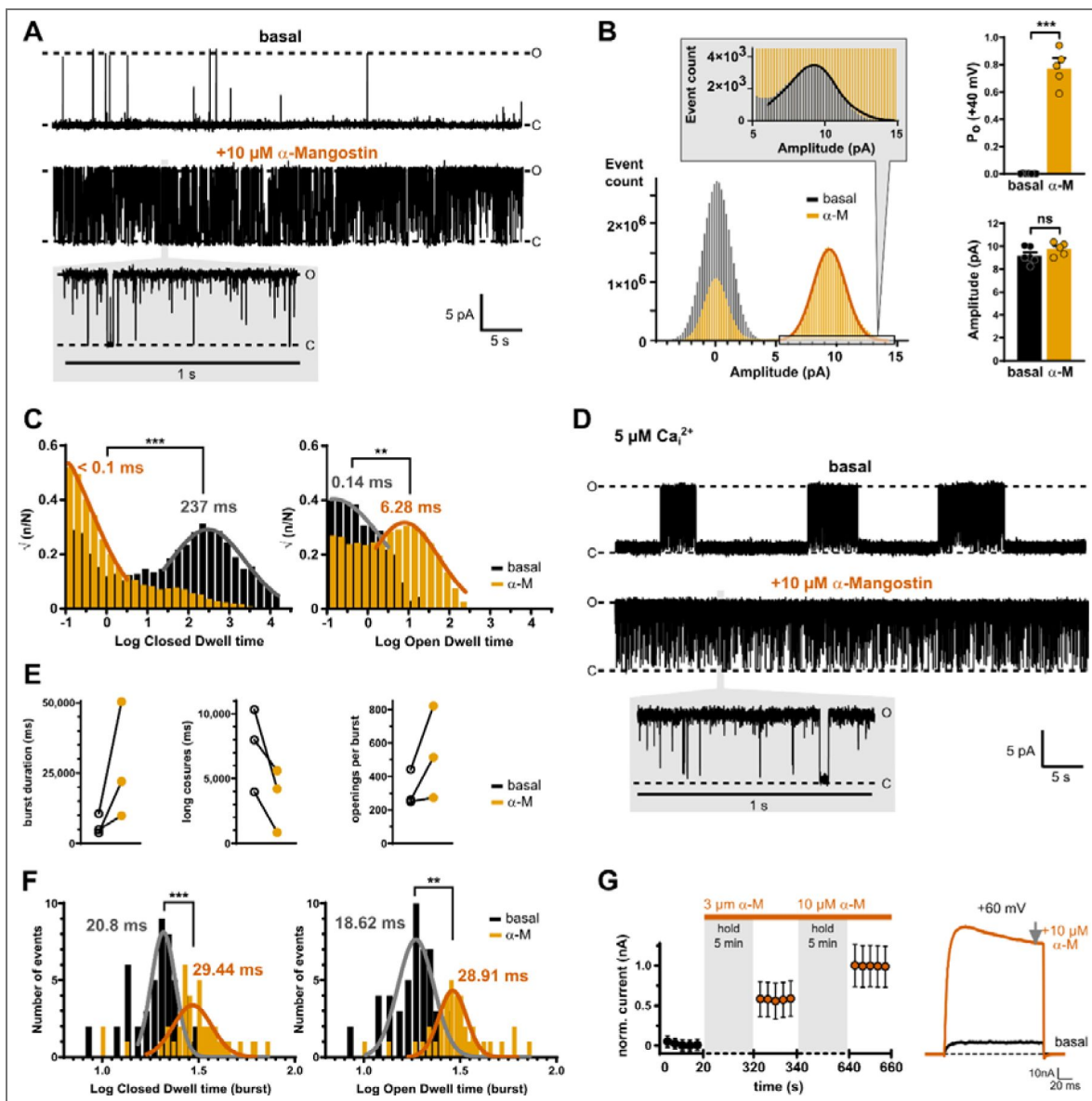


Figure 3. Activation mechanism of α-Mangostin.

(A) Exemplary current traces of a single BKα channel in the basal state and after activation by 10 μM α-Mangostin (1 min recordings; inset shows 1 s; O and C denote open and closed levels). Single-channel currents were recorded at +40 mV in inside-out patches from transiently transfected HEK293 cells in symmetrical potassium gradients with 100 mM free Ca_i^{2+} ($n=4-7$). (B) All-points histograms with Gaussian fits for the basal and α-Mangostin activated state, and bar graphs of the derived mean $\pm$ SEM open probabilities (P_o) and amplitudes. The inset magnifies the open peak in the basal state. (C) Closed and open dwell time histograms with fits for channels in the basal and in the α-Mangostin activated state derived after event detection in single-channel measurements as shown in (A). (D) Exemplary current traces of a single BKα channel recorded as in (A), but with 5 μM free Ca_i^{2+} . (E) Burst duration, duration of long closed times, and number of openings per burst derived from burst analysis ($n=3$ patches). (F) Closed and open dwell time distribution within bursts with fits for channels in the basal and in the α-Mangostin activated state. (G) Normalized mean $\pm$ SEM currents of BKα channels before and after application of 3 μM or 10 μM α-Mangostin in the closed state. Channels were held closed at -80 mV and only very shortly pulsed to +60 mV after 5 min incubation to assess the current size/activation state of the first and the following pulses. Measurements were done in transiently transfected HEK293 cells in whole-cell mode in physiological potassium gradients with 100 nM Ca_i^{2+} as shown by the representative current traces to the right and steady-state currents were analyzed (grey arrow). Data and statistics see Table S7-S8.

below the SF near the NCA binding site (Li and Aldrich, 2004 [\[1\]](#); Piechotta *et al.*, 2011 [\[2\]](#); Fan *et al.*, 2023 [\[3\]](#)). In a competition experiment, we recorded dose-response relationships and determined the IC_{50} value for Tetrapentylammonium (TPA) with and without preactivation by 10 μM α -Mangostin. The TPA IC_{50} value increased ≈ 9 -fold from $0.72 \pm 0.64 \mu M$ to $6.33 \pm 0.86 \mu M$ in the presence of α -Mangostin (Fig. S3B [\[4\]](#)), showing that α -Mangostin was present in the pore and hindered the access of TPA to its binding site. Finally, cysteine scanning mutagenesis of residues in M2 and M4 revealed that mainly G181, I182, P182 on M2 and G308 in M4 were involved, while most M4 residues tested had intermediary effects (Fig. S3C [\[5\]](#)), suggesting that the binding region of α -Mangostin between M2 and M4 overlaps with the NCA binding site identified with the help of BL-1249, but is not completely identical (Fig. S3D [\[6\]](#)).

We next probed if α -Mangostin accesses the pore cavity of BK α channels, as in TREK channels, or binds elsewhere in the protein. We conducted a competition experiment as above and measured the dose-dependent inhibition by 0.1, 1 and 10 μM Tetrahexylammonium (THexA; Fig. 4A [\[7\]](#)). In the presence of α -Mangostin, the inhibition by THexA was clearly reduced, and the apparent THexA affinity (as estimated IC_{50}) was steeply decreased ≈ 21 -fold from $77.51 \pm 5.53 nM$ to $1.64 \pm 0.58 \mu M$, showing that the presence of α -Mangostin hindered the access of THexA to its binding site. This is consistent with previous findings that GoSlo-RS-5-6, which binds in the BK pore, also competes with THexA (Schewe *et al.*, 2019 [\[8\]](#)). In contrast, the presence of BC5, an activator shown to bind at the interface between the transmembrane and intracellular domains (Zhang *et al.*, 2022 [\[9\]](#)), did not interfere with THexA binding (estimated THexA IC_{50} $45.60 \pm 8.05 nM$; Fig. 4A [\[7\]](#)).

At least one hydroxyl group of α -Mangostin is predicted to be negatively charged in physiological pH (National Center for Biotechnology Information, 2025 [\[10\]](#)), therefore we tested the impact of solution pH on the activation potency. Indeed, the shift in $V_{1/2}$ induced by 10 μM α -Mangostin was pH-dependent (Fig. 4C [\[11\]](#), S5). In pH 6, the shift was reduced to $34.71 \pm 5.63 mV$, while it increased to $75.97 \pm 2.02 mV$ in pH 8.5, indicating that the negative charge may be critical for effective activation and α -Mangostin might resemble an NCA-like compound.

In addition, we used molecular docking to determine the location of a possible binding site in the BK α inner pore in the Ca^{2+} -bound (presumably open) and the Ca^{2+} -free (presumably closed) structure (Fig. 4B [\[12\]](#), S4). We analyzed 20 poses which were all located at the cavity wall in middle S6 between residues I308 and V319, in a pocket below the SF, without obstructing the central passageway for K^+ ions (Fig. S4 [\[13\]](#)). In the Ca^{2+} -free and Ca^{2+} -bound structures, three and five poses were clustered with the lowest binding energies of $-8.58 kcal mol^{-1}$ and $-8.64 kcal mol^{-1}$, respectively. For the best pose in the Ca^{2+} -free state, possible molecular interactions were predicted for the residues I308, L312, F315 and A316, and α -Mangostin was wedged between S6 segments of adjacent helices, while in the Ca^{2+} -bound state where S6 moves upwards, I308 was buried and interactions shifted more to A316, resulting in a more horizontal molecule position (Fig. S4 [\[13\]](#)).

To functionally investigate the predicted binding site, we mutated residues in the pore-lining S6 helix with side chains facing into the cavity. Many substitutions in this region affect gating of BK channels and are involved in intersubunit S6-S6 contact (Wu *et al.*, 2009 [\[14\]](#); Zhou, Xia and Lingle, 2011 [\[15\]](#)). Therefore, we chose only substitutions that were most WT-like with respect to Ca^{2+} and voltage sensitivity, i.e., I308A, L312M, A316P for the hits from the docking, and additionally S317R and Y318S as internal controls that were not predicted to be part of the binding site (Chen, Yan and Aldrich, 2014 [\[16\]](#)). We first assessed the shift in voltage activation induced by 10 μM α -Mangostin (Fig. 4D, E [\[17\]](#)). $V_{1/2}$ shifts of Y318S and S317R were not or only moderately different compared to the wildtype channel ($41.46 \pm 5.17 mV$ and $34.33 \pm 3.37 mV$). In contrast, the three mutants I308A, L312M, and in particular A316P had a markedly reduced shift in $V_{1/2}$ ($19.97 \pm 3.12 mV$, $27.89 \pm 5.42 mV$, and $4.56 \pm 1.23 mV$, respectively). To ensure that the almost absent shift in A316P was not caused by a general disruption of allosteric signal transduction, we also included A316G, which showed a reduction in $V_{1/2}$ comparable to I308A and L312M ($23.57 \pm 2.05 mV$).

Furthermore, we tested GoSlo-SR-5-6 as alternative activator, which binds in the same pore region involving A316, and was shown to also compete with THexA (Schewe *et al.*, 2019 [\[8\]](#); Zhang *et al.*, 2022 [\[9\]](#)). Like for α -Mangostin, I308A and A316P reduced the shift in $V_{1/2}$ induced by 1 μM GoSlo-SR-

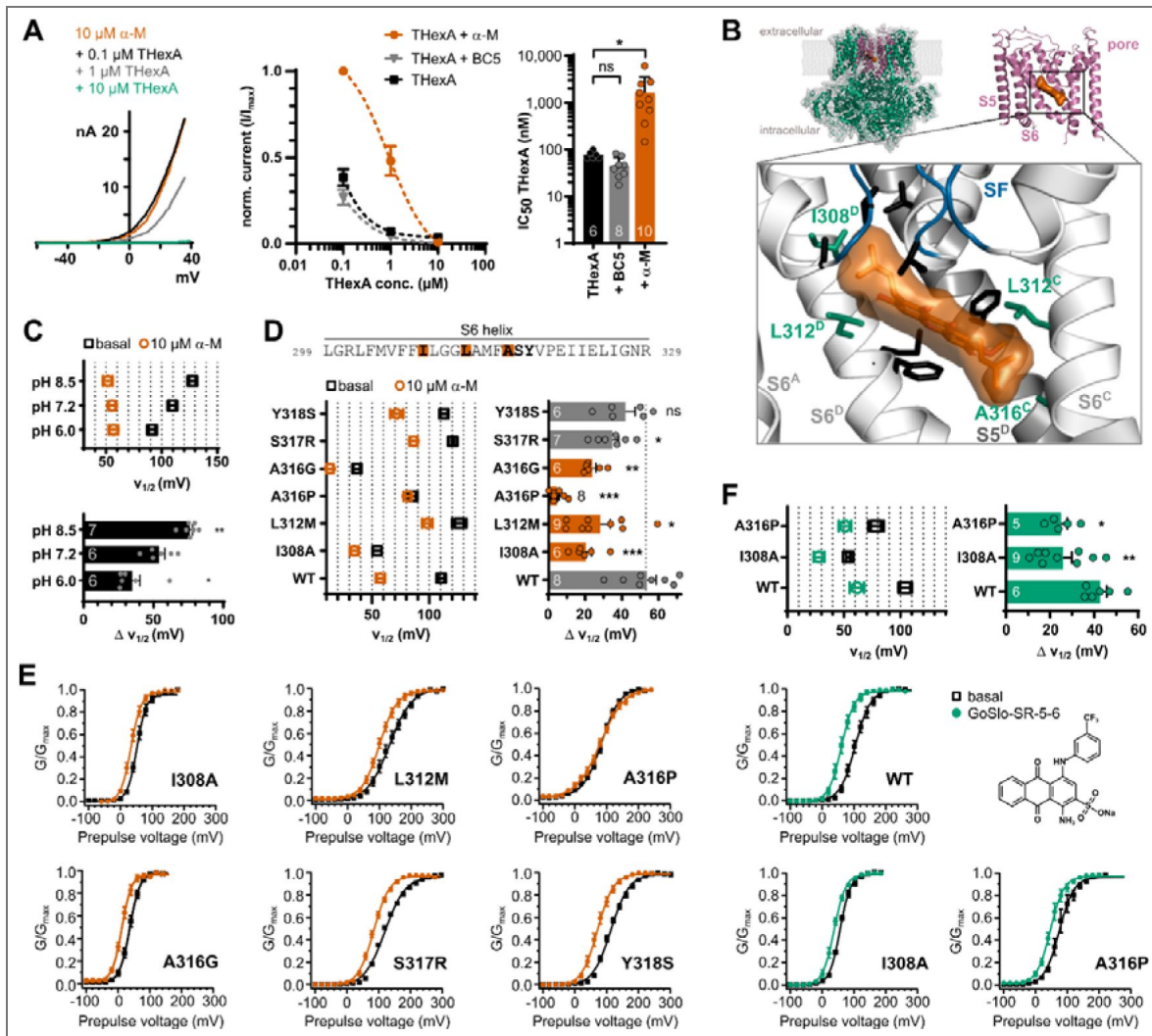


Figure 4. Investigation of the binding site of α -Mangostin in BK α channels.

(A) Competition experiment showing a reduction of the block caused by THexA in the presence of α -Mangostin. Left, representative current traces for different THexA concentrations in the presence of 10 μM α -Mangostin; middle, competition experiment analysis showing the relative current of BK α channels in different THexA concentrations in the absence and in the presence of 10 μM α -Mangostin or 100 μM BC5, which does not bind in the pore; and right, estimated IC₅₀ values of THexA alone and in the presence of α -Mangostin or BC5. Whole-cell currents were recorded from transiently transfected HEK293 cells with a ramp protocol (-100 mV to +50 mV) in a physiological potassium gradient with 100 nM free Ca_i²⁺ and data are shown as mean $\pm$ SEM at +40 mV. (B) Molecular docking of α -Mangostin to the human BK channel structure (PDB ID 6v3g, Ca²⁺-free state). The full-length structure (green) was reduced to the inner pore region (pink) for the docking, and the zoom-in shows the best pose for α -Mangostin with interacting residues in stick representation; green residues mark hits from the following functional assay. Protein chain B was removed for clarity. See Fig. S4 for the Ca²⁺-bound state. (C) Voltage of half-maximal activation ($V_{1/2}$) before and after activation by 10 μM α -Mangostin in different pH and the resulting shifts in $V_{1/2}$ ($\Delta V_{1/2}$). The pH was changed intra- and extracellularly. (D) Voltage of half-maximal activation ($V_{1/2}$) before and after activation by 10 μM α -Mangostin, and the resulting shifts in $V_{1/2}$ ($\Delta V_{1/2}$). (E) GV-relationships for the six BK α mutants in the S6 segment. (F) Voltage of half-maximal activation ($V_{1/2}$) before and after activation by 1 μM GoSlo-SR-5-6, the resulting shifts in $V_{1/2}$ ($\Delta V_{1/2}$), and the GV-relationships for the wildtype and two BK α mutants. GV-relationships were measured as in Fig. 2, and all data represent mean $\pm$ SEM. Data and statistics see Tables S11 – S14.

5-6 from 42.71 ± 3.05 mV to 25.89 ± 4.05 and 25.0 ± 2.8 mV; however, A316P did not abolish GoSlo-SR-5-6 activation, showing that the complete loss of activation for α -Mangostin was likely caused primarily by a loss of binding and not by an inference of the mutation with the drug transduction mechanism (Fig. 4F). Hence, we conclude that α -Mangostin binds to and activates BK α channels via the upper S6 segment, critically involving residues I308, L312, and A316.

α -Mangostin activation in BK- Ca_v nanodomains

In resting smooth muscle cells, there is no BK channel activation within their physiological voltage range. However, in native tissue, BK channels are organized in nanodomains together with Ca_v channels that can generate an increase in Ca^{2+} concentration in the range of several orders of magnitude in their vicinity, allowing BK channels to open (Shah, Guan and Yan, 2022). This BK- Ca_v complex can be restored in heterologous expression systems, and, as the intracellular solution contains 5 mM EGTA to suppress an overall increase of Ca^{2+} , BK channels will activate only when nearby Ca_v channels cause a high local Ca^{2+} concentration (Berkefeld *et al.*, 2006; Berkefeld and Fakler, 2008). BK α / β 1 channels expressed alone therefore produced virtually no current upon stepwise depolarization to +50 mV (Fig. 5A, B), but could be activated by α -Mangostin (Fig. 5A, B). In BK α / β 1- Ca_v complexes, Ca^{2+} inward currents were elicited, which allowed BK α / β 1 channels to open in a more negative voltage range, and these BK α / β 1 currents were further enhanced by 10 μM α -Mangostin (Fig. 5A, B). The α -Mangostin-activated current (as difference between BK α / β 1- Ca_v currents before and after activation) was activated at more negative potentials when BK α / β 1 and Ca_v channels were coexpressed than when BK α / β 1 channels were expressed alone (Fig. 5B). Hence, α -Mangostin could potentiate BK α / β 1 channel currents upon local Ca_i^{2+} increase through nearby Ca_v -channels, as it could occur upon a Ca_i^{2+} spike in a cell.

α -Mangostin relaxes vascular tissue in aortic preparations via BK channels

Cardiovascular benefits of Mangostins could arise from vasodilatory effects leading to a reduction of blood pressure. Years ago, a study reported that γ -Mangostin, in which the methoxy group of α -Mangostin is exchanged for a hydroxyl group, induced relaxation of rat aortic rings, but no molecular target was found (Tep-Areenan and Suksamrarn, 2012). We sought to demonstrate that Mangostin-induced vasorelaxation of native vascular tissue is indeed mediated by BK channels. Aortic preparations from mice were equilibrated at a contraction force of 2-3 mN and subsequently precontracted half-maximally with 100 nM noradrenaline. The application of 10 μM α -Mangostin quickly and efficiently induced relaxation, while relaxation was absent or greatly attenuated when the specific BK channel inhibitor Iberiotoxin (IbTx) was applied for 5-6 minutes before α -Mangostin was administered (Fig. 5C). In total, α -Mangostin reduced the normalized contraction force of the preparations by more than 85 % (to 0.16 ± 0.08) compared to the precontracted state, while it did not change after IbTx application alone (1.04 ± 0.002) or after preblock of BK channels (0.78 ± 0.12), proving that BK channels must mediate the vasodilative effect seen upon Mangostin treatment.

Discussion

We provide mechanistic insight into how a natural xanthone compound modulates BK channels, which are major players in vascular function. α -Mangostin activates BK channels by shifting their voltage sensitivity and slowing deactivation, with lesser effects on activation and marked effects on Ca^{2+} sensitivity. We show that binding of α -Mangostin is state-independent, and we provide evidence for a binding site in the gating-sensitive S6 segment.

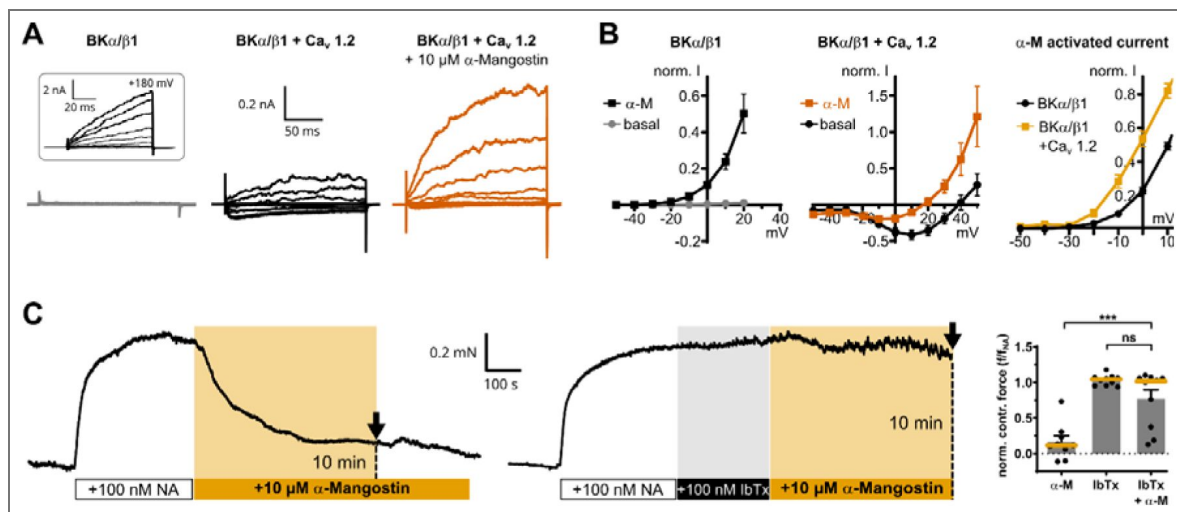


Figure 5. α -Mangostin activation of BK channels in physiological settings.

(A) Representative whole-cell current traces of BK α / β 1 channels alone and BK α / β 1 coexpressed with Ca $_v$ 1.2 channels before and after application of 10 μ M α -Mangostin. Currents were measured in Ca $_i$ ²⁺-free conditions in a physiological potassium gradient with a family of voltage steps from -50 mV to +50 mV in 10 mV increments. The inset shows voltage activation with a family protocol up to +200 mV to show the presence of BK α / β 1 channels. **(B)** Currents of BK α / β 1 channels and BK α / β 1 – Ca $_v$ complexes before and after application of 10 μ M α -Mangostin plotted against voltage. The last panel shows the α -Mangostin-activated currents for the range -50 to 10 mV obtained by subtracting the current before α -Mangostin application from the current after application for each potential (mean $\pm$ SEM, n=8-11 for each condition). **(C)** Representative contraction force recordings of aortic preparations from mice. 10 μ M α -Mangostin were either applied directly to aortic preparations precontracted with 100 nM Noradrenaline (NA; top), or the precontracted preparations were incubated with 100 nM Iberiotoxin (IbTx) before α -Mangostin application and the contraction force was analyzed 10 min after α -Mangostin addition (dotted lines in recordings). The bar graph shows the normalized contraction force of preparations as mean $\pm$ SEM together with the median (orange). Data and statistics see Table S16. DMSO controls are shown in Fig. S6.

Mangostin binds to closed and open channels and promotes the open state

BK channel P_o is controlled by Ca^{2+} , voltage, a shift of the closed-open equilibrium of the pore domain, or a combination of those stimuli (Cui, Cox and Aldrich, 1997 [\[1\]](#)). Mangostin shifted the voltage activation curve to more negative potentials, meaning an increased open probability at negative voltages. The slope of the GV-relationship was not changed neither in homomeric BK α nor in the smooth muscle BK α / β 1 channels, indicating that voltage sensitivity itself was not directly affected by α -Mangostin. Shifts in the $V_{1/2}$ values induced by α -Mangostin were present over a wide range of Ca^{2+} concentrations (0.1, 1 and 10 μ M), which implies that the closed-open equilibrium is shifted towards the open state by a mechanism distinct from Ca_i^{2+} sensing, or that at least the Ca_i^{2+} sensing mechanism is not likely to contribute significantly to α -Mangostin activation. The higher shift observed in 1 μ M Ca_i^{2+} may reflect the steep Ca_i^{2+} -dependence of the closed-open equilibrium (Cui, Cox and Aldrich, 1997 [\[1\]](#)) and the allosteric coupling of voltage and Ca^{2+} signals (Horrigan and Aldrich, 2002 [\[2\]](#); Magleby, 2003 [\[3\]](#); Clay, 2017 [\[4\]](#)) which are effective in this concentration range, and which may lead to a higher apparent activation when voltage activation is facilitated by Ca^{2+} (Sun and Horrigan, 2022 [\[5\]](#)).

The activation can be mainly explained by the marked slowing of the deactivation time constants. Consistently, the mean open dwell time was increased in the single channel measurements and brief closings within bursts were reduced, while we observed no change in amplitude with and without α -Mangostin (corresponding to a conductance of $\approx$ 240 pS). The impact on open as well as closed dwell time distributions and the instantaneous activation of macroscopic currents from the closed state suggest that the molecule binds state-independently.

In the presence of the β 1 subunit, as in vascular BK channels (Knaus *et al.*, 1994 [\[6\]](#)), the activation characteristics were enhanced. The activation time course was little affected in BK α channels, but in BK α / β 1 channels an acceleration of the activation caused by α -Mangostin became more prominent, counteracting the slow activation kinetics brought by the β 1 subunit (McManus *et al.*, 1995 [\[7\]](#)). Additionally, the slow deactivation time course in the presence of the β 1 subunit was further slowed down remarkably. As the affinity (as estimated EC_{50}) was very similar and the binding site is located in the S6 segment in the inner pore, the stronger shift in the $V_{1/2}$ value is likely not due to changes in binding, but reflects the enhanced shift of the closed-open equilibrium toward the open state conferred by the β 1 subunit. Such β -subunit dependent effects have also been reported for other modulators, e.g., DHS-I (McManus *et al.*, 1993 [\[8\]](#); Giangiacomo *et al.*, 1998 [\[9\]](#)), DHA (Hoshi, Tian, *et al.*, 2013 [\[10\]](#)), arachidonic acid (Martín *et al.*, 2021 [\[11\]](#)), 17 β -Estradiol (Valverde *et al.*, 1999 [\[12\]](#)), and some GoSlo compounds (Large *et al.*, 2015 [\[13\]](#)).

The binding site of α -Mangostin includes residues in the gating-sensitive S6 segment

We demonstrated the presence of α -Mangostin in the channel pore by competition for binding with quaternary ammonium ions known to bind below the SF (Li and Aldrich, 2004 [\[14\]](#); Fan *et al.*, 2023 [\[15\]](#)), and, predicted by molecular docking to the Ca^{2+} -free and the Ca^{2+} -bound state, identified residues in the S6 segment that are part of the binding site.

α -Mangostin binds in the cleft formed by the S6 helices of adjacent subunits, and substitution of the nonpolar residues I308, L312, and especially A316 most strongly reduced the shift in $V_{1/2}$ induced by α -Mangostin. All residues are known to participate in gating: they are part of a hydrophobic network within S6 that acts as hinge upon channel opening, and polar substitutions strongly favor the open state (Zhou, Xia and Lingle, 2011 [\[16\]](#); Chen, Yan and Aldrich, 2014 [\[17\]](#); Hite, Tao and MacKinnon, 2017 [\[18\]](#)). L312 has further been shown to stabilize intersubunit interaction with F315, whose destruction opens the channel (Wu *et al.*, 2009 [\[19\]](#)). In the open state, S6 is thought to move upwards, as seen in the Ca^{2+} -bound cryo-EM structure (Kallure *et al.*, 2023 [\[20\]](#); Yamanouchi

et al., 2023 [\[1\]](#); Redhardt, Raunser and Raisch, 2024 [\[2\]](#)), thereby burying the I308 sidechain and diminishing the dimension of the upper cleft. Therefore, the contribution of I308 to α -Mangostin activation is likely allosteric in nature, and the pose of the molecule changes with the S6 position.

The α -Mangostin binding region overlaps with the reported binding site of the NCA GoSlo-SR-5-6, where L312, A316, S317 and V319 are involved (Webb *et al.*, 2015 [\[3\]](#); Schewe *et al.*, 2019 [\[4\]](#)), but it is not identical. When we functionally investigated the α -Mangostin binding site close to the fenestration in TREK-1 channels, we also found only a partial overlap with key residues involved in the NCA binding site. Interaction with P183 and L304 was consistent with an earlier molecular docking (Kim *et al.*, 2023 [\[5\]](#)), but the strongest reduction of activation was found for residues in M2 rather than M4.

Mechanism of Mangostin activation

We expect that small structural or electrostatic changes in this gating-sensitive region strongly affect the closed–open equilibrium (Wu *et al.*, 2009 [\[6\]](#); Zhou, Xia and Lingle, 2011 [\[7\]](#); Chen, Yan and Aldrich, 2014 [\[8\]](#)). Previous studies suggest several alternative mechanisms by which α -Mangostin could induce channel opening. We have shown that NCAs bind to a region below the selectivity filter (SF) in several K^+ channels, such as K_{2p} (e.g., TREK), hERG, or BK channels to induce channel opening (Schewe *et al.*, 2019 [\[4\]](#)). We speculated that the negative charge common to these activators might alter SF ion occupancy via an electrostatic mechanism, thereby promoting SF opening. Our finding that the potency of α -Mangostin depends on pH suggests that a negative charge may also be critical for its activation mechanism. However, the importance of the SF for BK gating remains unresolved. Furthermore, α -Mangostin also did not increase the single-channel conductance as seen for NCAs in TREK-2 channels, which are known to be gated at the SF (Schewe *et al.*, 2019 [\[4\]](#)). Therefore, the contribution of the SF to the Mangostin effect requires further investigation. Alternatively, α -Mangostin binding to the gating-critical S6 region could induce gating at a putative lower gate; however, such a gate is currently speculative, as the available BK structures lack clear evidence for a lower gate (but the final closed state may still be elusive) (Hite, Tao and MacKinnon, 2017 [\[9\]](#); Tao, Hite and MacKinnon, 2017 [\[10\]](#)).

In addition to the concepts of an SF gate or a lower gate, hydrophobic gating has been proposed as an alternative gating mechanism in BK channels. MD simulations suggest that the pore of metal-free (presumably closed) BK structures undergoes dewetting, whereas the pore of metal-bound (presumably open) structures remains hydrated (Jia *et al.*, 2018 [\[11\]](#); Gu and de Groot, 2023 [\[12\]](#)). Thus, binding of negatively charged α -Mangostin might prevent or antagonize dewetting of the hydrophobic cavity and thereby stabilize the conductive pore state, probably with substantial involvement of I308. Indeed, a recent MD study suggested that the NCA NS11021, a smooth muscle relaxant with a biarylthiourea structure, can enter the dewetted BK pore to promote hydration (Rockman, Vouga and Rothberg, 2020 [\[13\]](#); Nordquist, Jia and Chen, 2024 [\[14\]](#)). However, the same MD simulations also indicated that NS11021 has no stable binding pose but engages in various hydrophobic interactions with pore-cavity residues. In contrast, our results suggest a more stable binding pose with specific interactions to a number of residues in the Ca^{2+} -free as well as in the Ca^{2+} -bound state, as inferred from our mutagenesis and docking data. Clearly, further work is required to elucidate the mechanism of BK channel gating and the exact way Mangostin and other NCAs modulate this process to promote channel activation.

Activation of BK channels in nanodomains and relaxation of aortic tissue

Our data demonstrate that α -Mangostin is able to potentiate Ca_v -induced currents, and that α -Mangostin-activated BK channels are responsible for relaxation of mouse vascular tissue, explaining the reduction of blood pressure reported in a recent animal study (Xu *et al.*, 2024 [\[15\]](#)).

We reconstructed nanodomains of BK and $Ca_v1.2$ channels, as e.g. present in arterioles (Berkefeld *et al.*, 2006 [\[16\]](#); Tykocki, Boerman and Jackson, 2017 [\[17\]](#)). In smooth muscle cells, BK activation is strictly coupled to changes in Ca_i^{2+} . Physiological Ca_i^{2+} concentrations lie between 100 nM in the

resting state, 0.4–1 μM upon activation of a Ca^{2+} mobilizing receptor (Savineau and Marthan, 2000; Hill-Eubanks *et al.*, 2011), and up to several ten micromolar (10–40 μM) when sparks form locally (ZhuGe *et al.*, 2002; Berkefeld, Fakler and Schulte, 2010). Transient outward currents through activated BK channels then act as negative feedback, repolarize the cell and terminate Ca^{2+} influx (Shah, Guan and Yan, 2022). We demonstrated that the α -Mangostin activated current is potentiated by the coupling to Ca_v channels. BK α / β 1 currents in the physiological voltage range of smooth muscle cells (–50 – –30 mV) would be very small in resting Ca^{2+} , while the P_0 would be already very high when exceeding 10 μM Ca^{2+} , leaving less room for activation. α -Mangostin would particularly impact potentiation of BK currents upon physiological Ca^{2+} elevation (i.e. between 1 and 10 μM) as well as in the physiological voltage range, and would synergistically lower the BK α / β 1 activation threshold to enhance the negative feedback mechanism.

Accordingly, our investigation of precontracted aortic tissue from mice showed a remarkable and robust relaxation after application of α -Mangostin. Such relaxation of aortic rings after application of α - or γ -Mangostin were reported before (Chairungsrikerd *et al.*, 1996, 1997; Tep-Areenan and Suksamrarn, 2012) but the exact molecular target remained inconsistent. Given the strong activation of BK channels and as the relaxation was absent upon preincubation with the selective inhibitor IbTx in our experiments, BK channels must represent the molecular target of Mangostins in vascular smooth muscle cells.

Interestingly, relaxation was also reported for other xanthone derivatives (Cheng and Kang, 1997; Wang *et al.*, 2002; Capettini *et al.*, 2009; Câmara *et al.*, 2010; Diniz *et al.*, 2013), which legitimates further research of substituted xanthenes as pharmacophore. Robust data of bioavailability and the existence and nature of active metabolites are currently lacking. A small study claimed that α -Mangostin was bioavailable in humans after supplement ingestion (Kondo *et al.*, 2009). Animal research analyzed metabolites and showed that maximal plasma concentrations are reached 1 hour after oral intake in mice (Ramaiya *et al.*, 2012; Petiwala *et al.*, 2014; Han *et al.*, 2015). However, no metabolites have been functionally investigated to help in optimization of the Mangostin pharmacophore; however, our finding that γ -Mangostin has a slightly higher potency in activating TREK-1 and BK α channels may be a starting point.

Our screen across different K^+ channel families revealed that other representatives of the $\text{K}_{2\text{P}}$ (TWIK-1 and TASK-3) were particularly inhibited by more than 60 %. This could also link other claimed benefits of Mangostin to their molecular targets. Importantly, hERG currents were unaffected, mitigating the cardiotoxicity risk upon consumption of Mangostin nutraceuticals. BK channel function is decreased in chronic conditions that are often associated with the lack of a balanced diet and age, such as metabolic syndrome, diabetes, and obesity (Tykocki, Boerman and Jackson, 2017). Consequences as increased vascular tone and hypertension, ultimately leading to reduced cardiovascular health and premature death, may therefore be addressed with Mangostin xanthenes as a potential new class of BK channel activators.

Methods and Materials

Substances

α -Mangostin, γ -Mangostin, TPA, THexA, MgATP, NaGTP and noradrenaline were purchased from Merck (Darmstadt, Germany). BC5 (Arg-4-methoxy-2-naphthylamine) was obtained from MP Biomedicals (Irvine, USA), and Iberiotoxin from Alomone Labs (Jerusalem, Israel). GoSlo-SR-5-6 (sodium 1-amino-4-((3-trifluoromethylphenyl)amino)-9,10-dioxo-9,10-dihydroanthracene-2-sulfonate) was provided by Mark Hollywood. Mangostane food supplement stating a content of 50 mg α -Mangostin per capsule was ordered online (Swanson.com). Aiming for a 10 mM stock solution with respect to α -Mangostin, 205 mg of the powder found in the capsules was dispersed in 5 ml DMSO, brought into solution by vortexing and ultrasonication, and filtered through a 0.45 μM PTFE membrane to remove remaining particulate matter. TPA, THexA, noradrenaline, BC5, and

Iberitoxin were prepared as 1:1000 stocks in H₂O, and the Mangostins and GoSlo-RS-5-6 in DMSO at 50 mM. Aliquots were stored at -20°C and diluted to the final concentration in extracellular solution. The final DMSO concentration did not exceed 0.025 %.

Molecular biology, cell culture and transfection

Coding sequences for channels and subunits listed in Table 1 [\[link\]](#) were subcloned in pcDNA3.1 (<https://www.addgene.org/vector-database/2097/> [\[link\]](#)) or pFAW vectors containing a CMV promoter for expression in cultured cells, or in the pBF *Xenopus* oocyte vector for low expression in the single channel experiments (see below). Amino acid substitutions were introduced by site-directed mutagenesis PCR according to the QuikChange protocol (Stratagene, La Jolla, USA) with custom primers containing the desired base exchange. All constructs were verified by Sanger sequencing. To enhance plasma membrane expression, the retrieval motif was removed in hTWIK-1 and hTWIK-2 channels by the substitutions I293A, I294A (Felicangeli *et al.*, 2010 [\[link\]](#)) and I289A, L290A (Bobak *et al.*, 2017 [\[link\]](#)), respectively, yielding hTWIK-1^m and hTWIK-2^m channels.

HEK293 cells (RRID:CVCL_0045 [\[link\]](#)) were cultivated in Dulbecco's Modified Eagles Medium (DMEM) supplemented with 10 % FCS and penicillin/streptomycin (100 U ml⁻¹/100 µg ml⁻¹) in 5 % CO₂ at 37 °C. For cultivation of CHO-K1 cells (RRID:CVCL_0214 [\[link\]](#)), the medium was additionally supplemented with non-essential amino acid mix (Gibco MEM-NEAA; Thermo Fisher, Schwerte, Germany) and 10 mM Hepes. Cell lines were regularly tested for mycoplasma contamination by PCR (Venor GEM OneStep, Minerva Biolabs).

HEK293 cells were transiently transfected with Lipofectamine 2000 (Invitrogen, Thermo Fisher, Schwerte, Germany) or FuGENE (Promega, Walldorf, Germany) as to the manufacturer's instructions and incubated overnight. pEYFP (<https://www.addgene.org/vector-database/2688/> [\[link\]](#)) was included as transfection marker. In BKα/β1 coexpressions, the β1 subunit was used in 10-fold excess to ensure a uniform channel population. Cells were trypsinized and seeded onto glass coverslips 2-3 h prior to electrophysiological experiments.

Electrophysiology

Voltage clamp experiments were conducted with a HEKA EPC10 amplifier (RRID:SCR_018399 [\[link\]](#)) controlled by Patchmaster Software (v2.78; HEKA Elektronik, Lambrecht, Germany). Standard measurements were done in transiently transfected HEK293 cells in the whole-cell configuration in physiological potassium gradients. Pipettes were pulled from thin-walled borosilicate glass, fire-polished, and had resistances of 1.4 – 2 MΩ. Series resistance was compensated to at least 70 %. The extracellular solution contained (in mM): 135 NaCl, 5 KCl, 2 CaCl₂, 2 MgCl₂, 10 D(+)-glucose, 10 Hepes (pH 7.3). The intracellular solution was (in mM): 140 KCl, 2 MgCl₂, 1 CaCl₂, 2.5 EGTA, 10 Hepes (pH 7.3), corresponding to ≈100 nM free Ca²⁺. For K_{ir} 1.1 and 2.1 channel recordings, 3 mM MgATP and 0.3 mM NaGTP were included. Where indicated, K⁺ was substituted for Cs⁺ to reduce the outward currents of BK channels. Cells were held at -80 mV and a voltage ramp from -100 mV to +60 mV of 1 s duration was applied every 5 s. For K_v channels a rectangle pulse was used as indicated in the figure legends.

The activation of BK channels was measured as conductance in symmetrical potassium solutions to ensure a unitary conductance across the voltage range. Cs⁺ was used as intracellular ion to reduce outward currents. The extracellular solution contained (in mM): 140 KCl, 2 CaCl₂, 2 MgCl₂, 10 D(+)-glucose, 10 Hepes (pH 7.3). The intracellular solution was (in mM): 140 CsCl, 2 MgCl₂, 1 CaCl₂, 2.5 EGTA, 10 Hepes (pH 7.3), corresponding to ≈100 nM free Ca²⁺ (calculated with WEBMAXC (<https://somapp.ucdmc.ucdavis.edu/pharmacology/bers/maxchelator/webmaxc/webmaxcS.htm> [\[link\]](#))). For recordings with 1 µM and 10 µM free Ca²⁺, CaCl₂ and EGTA were raised to 2 mM / 2.2 mM and 2 mM / 4.04 mM, respectively. Cells were held at negative holding potentials and currents were elicited with a family of rectangle voltage pulses in 20 mV increments to positive potentials as needed for saturation of the tail currents, followed by a repolarization step to elicit the tail currents as detailed in the figure legends. The currents were analyzed with Fitmaster (HEKA

Table 1. Channels and subunits used in this study

Name	Organism	Gene name	Genbank Acc. No.
hTREK-1b	<i>Homo sapiens</i>	KCNK2	NM_001017425.2
hTALK-1	<i>Homo sapiens</i>	KCNK16	NM_032115.3
hTRESK	<i>Homo sapiens</i>	KCNK18	NM_181840.1
hTWIK-1 ^m	<i>Homo sapiens</i>	KCNK1	NM_002245.3
hTWIK-2 ^m	<i>Homo sapiens</i>	KCNK6	NM_004823.1
hTASK-3	<i>Homo sapiens</i>	KCNK9	NM_001282534.1
hTHIK-1	<i>Homo sapiens</i>	KCNK13	NM_022054.3
mBKα (Slo1.1) ¹	<i>Mus musculus</i>	Kcnma1	NM_001014797.3
mβ1	<i>Mus musculus</i>	Kcnmb1	NM_031169.4
hK _v 1.1	<i>Homo sapiens</i>	KCNA1	NM_000217.3
hK _v 1.3	<i>Homo sapiens</i>	KCNA3	NM_002232.5
hK _v 11.1 (hERG1b)	<i>Homo sapiens</i>	KCNH2	AJ512214.1
hK _{ir} 1.1 (ROMK)	<i>Homo sapiens</i>	KCNJ1	NM_000220.4
mK _{ir} 2.1	<i>Mus musculus</i>	KCNJ2	NM_008425.4
Ca _v 1.2:			
Ca _v α 1C	<i>Rattus norvegicus</i>	Cacna1c	M67515.1
Ca _v β1	<i>Rattus norvegicus</i>	Cacnb1	NM_017346.1
Ca _v α2δ1	<i>Rattus norvegicus</i>	Cacna2d1	AF286488

¹ G/S-rich N-terminus removed, corresponding to residues 65-1236 of the native channel

Elektronik, Lambrecht, Germany; *RRID:SCR_016233* [\[link\]](#)), and conductance-voltage (G-V) relationships were generated from normalized tail currents and fit to a standard Boltzmann relationship (1) to obtain $V_{1/2}$ values. Individual fits were then averaged for mean $V_{1/2}$ values.

$$\frac{G}{G_{max}} = \frac{1}{1 + e^{\frac{(V - V_{1/2})}{k}}}$$

where G_{max} is the maximal tail current, $V_{1/2}$ is the voltage of half-maximal activation of the current, and k is the slope factor.

Activation in different pH was measured using the following solutions: Extracellular solution pH 8.5 contained (in mM): 140 KCl, 2 CaCl₂, 2 MgCl₂, 10 D(+)-glucose, 10 Ampso; intracellular solution pH 8.5 contained (in mM): 140 CsCl₂, 10 Ampso, 2 MgCl₂, 1 CaCl₂, 2.05 EGTA. Extracellular solution pH 6 was (in mM): 140 KCl, 2 CaCl₂, 2 MgCl₂, 10 D(+)-glucose, 10 MES; intracellular solution pH 6 contained (in mM): 140 CsCl₂, 10 MES, 2 MgCl₂, 0.5 CaCl₂, 11 EDTA. The pH was changed intra- and extracellularly.

In TPA competition experiments for TREK-1 channels a dose-response relationship in the absence and presence of α -Mangostin was recorded and the IC₅₀ was obtained from a Hill fit. The THexA competition experiments for BK α channels were conducted as 3-point determinations using 0.1 μ M and 1 μ M THexA as concentrations close to half-maximal inhibition without and with α -Mangostin and 10 μ M THexA for full block. To estimate the IC₅₀, a simplified 4-parameter logistic regression was used, assuming $a = 0$ and $b = 1$.

$$y = d + \frac{a-d}{1 + (\frac{x}{c})^b} \rightarrow c = \frac{x}{(\frac{d}{y} - 1)^{1/b}}$$

where a : lower asymptote, b : slope, c : IC₅₀, d : upper asymptote, x : THexA concentration, and y : current.

Single-channel recordings

Single channels were recorded at room temperature from excised inside-out patches of HEK293 cells transiently transfected with BK α in a continuous voltage protocol at +40 mV. To increase the chances of obtaining patches with only one channel, a pBF vector with a suboptimal beta globin promotor yielding only low expression was used. Pipette resistances were 12-15 M Ω , and symmetrical intra- and extracellular solution contained (in mM) 140 KCl, 2 MgCl₂, 1 CaCl₂, 2.5 EGTA, 10 Hepes (pH 7.3 with KOH), corresponding to $\approx$ 100 nM free Ca²⁺, or 140 KCl, 2 MgCl₂, 2.36 CaCl₂, 2.4 EGTA, 10 Hepes (pH 7.3 with KOH), corresponding to $\approx$ 5 μ M free Ca²⁺. Currents were recorded at a sampling rate of 100 kHz with a final bandwidth f_c of 7.4 kHz using an EPC10 amplifier (*RRID:SCR_018399* [\[link\]](#)) controlled by Patchmaster software (v2.78; HEKA Elektronik, Lambrecht, Germany; *RRID:SCR_000034* [\[link\]](#)). Traces were analyzed in Clampfit (v11.2.2.17; Molecular Devices, San Jose, USA). The filter rise time $\tau_r = \frac{0.3321}{f_c}$ was 45 μ s, and openings $> 2 T_r$ were fitted (Colquhoun and Sigworth, 1995 [\[link\]](#)). Histograms were calculated in Clampfit and fitted with a Gauss function to obtain amplitudes and dwell times using Prism (v8.4.3; GraphPad Software Inc., San Diego, USA; *RRID:SCR_002798* [\[link\]](#)). Single channel conductance was calculated from $g = \frac{I}{(V_m - E_{K^+})}$, where g , conductance; I , single channel current; V_m , membrane voltage; E_{K^+} , K⁺ equilibrium potential. Bursts were detected using a minimum closed interval of 200 ms. Bursts with a single opening (0 ms intraburst interval) were included in the calculation of mean closed times within a burst. Means were calculated per patch. The fits of kinetic distribution data are descriptive as data was pooled.

Reconstitution of nanodomains

BK α / β 1 – Ca ν complexes were restored as described by (Berkefeld *et al.*, 2006 [\[link\]](#)) with the following modification: For nanodomain formation of Ca ν 1.2 and BK α / β 1, CHO-K1 cells were seeded onto coverslips and microinjected (InjectMan 4, Eppendorf, Hamburg, Germany) with a DNA mixture of Ca $\nu\alpha$ 1C, Ca $\nu\beta$ 1, Ca $\nu\alpha$ 2 δ 1, BK α , β 1 and EYFP in the ratio 10:10:10:10:100:1 to ensure the presence of all subunits. Cells were incubated overnight and currents were recorded from EYFP-positive

cells 14-18 h after injection. Cells were measured in a physiological potassium gradient with a voltage step protocol from -50 mV to +60 mV in 10 mV increments. The extracellular solution contained (in mM): 135 NaCl, 5 KCl, 2 CaCl₂, 2 MgCl₂, 10 D(+)-glucose, 10 Hepes (pH 7.3). The intracellular solution was (in mM): 140 KCl, 2 MgCl₂, 5 EGTA, 10 Hepes (pH 7.3). α-Mangostin was added to the bath solution at a final concentration of 10 μM.

Molecular Docking

The input structures for molecular docking were generated by truncating the structure of the human BKα channel in the Ca²⁺-free and Ca²⁺-bound state (PDB ID 6v3g, 6v38) to only the pore region (res T229-R329) using PyMOL (Schrodinger LLC, 2015 [DOI](#); *RRID:SCR_000305* [DOI](#)). Docking was performed with AutoDock4 (v4.2.6; *RRID:SCR_012746* [DOI](#)) using MGL Tools (v1.5.7) (Morris *et al.*, 2009 [DOI](#)). A run with lower resolution grid was done and after α-Mangostin was seen in the pore, the docking was repeated with a gridbox restricted to the cavity below the SF. Clustering and interactions were analyzed with the built-in functions of MGL Tools.

Aortic smooth muscle preparations

1 female and 7 male CD1/CHR2 mice (*RRID:IMSR_CRL:022* [DOI](#)) at the age of 5-6 months were randomly chosen and killed by decapitation following isoflurane-narcosis and the aorta was removed. Inclusion/exclusion criteria did not apply and all animals were used for further study. The aortic wall was dissected and tissue strips of ~15x2 mm were obtained using the spiral cut technique (Peiper and Schmidt, 1972 [DOI](#)). Aortic preparations were tethered to glass holders and transferred to temperature-controlled 37 °C bath chambers filled with Krebs solution (pH 7.4; in mM: 112 NaCl, 4.7 KCl, 2.5 CaCl₂, 1.2 MgCl₂, 1.2 KH₂PO₄, 25 NaHCO₃ and 11.5 Glucose). The bath solution was bubbled with 95 % O₂/ 5 % CO₂. The contraction force was recorded with mechanoelectric transducers (Fort100, WPI, Sarasota, USA), connected to a PowerLab controlled by LabChart (AD Instruments, Mannheim, Germany; *RRID:SCR_018833* [DOI](#)). A prestrain of 2.5-3.9 mN was applied, depending on the thickness of preparations, and strips equilibrated for ca. 45 minutes. The NA concentration necessary to achieve half-maximal contraction was determined by measuring a dose-response relationship and 100 nM NA were subsequently used for precontraction. After a plateau was reached (2-4 minutes), α-Mangostin and IbTx were diluted into the bath solution to their final concentrations of 10 μM (final DMSO concentration 0.1 %) and 100 nM, respectively, and the contraction force was recorded for ca. 20 minutes. Recordings were analyzed with LabChart.

Data analysis and statistics

All data are given as mean ± SEM. Mean currents were analyzed with Patchmaster (HEKA Elektronik, Lambrecht, Germany; *RRID:SCR_000034* [DOI](#)). Fitmaster (HEKA Elektronik, Lambrecht, Germany; *RRID:SCR_016233* [DOI](#)) was used to generate GV-relationships and fits with the Boltzmann function, and for the monoexponential fits to obtain activation and deactivation time constants. EC₅₀ / IC₅₀ values were obtained by fitting a Hill equation to the dose-response data using IgorPro (v6.3.7; WaveMetrics, Portland, USA; *RRID:SCR_000325* [DOI](#)) or Prism (v8.4.3; Graphpad Software Inc., San Diego, USA; *RRID:SCR_002798* [DOI](#)).

$$I_0 + \frac{I_{\max} - I_0}{1 + \left(\frac{c_{1/2}}{c}\right)^h}$$

With I₀, basal current; I_{max}, activated current; c, concentration; c_{1/2} EC₅₀ / IC₅₀; h, Hill coefficient. Statistical analysis was conducted with the built-in functions of Prism. If data were not normally distributed, nonparametric tests were used. Variances were compared with F-tests. Two means were then compared with a paired or unpaired two-tailed t-test. Three or more groups were compared using one-way ANOVA when variances and standard deviation were considered equal by Bartlett's test, otherwise the Brown-Forsythe and Welch ANOVA was applied. Dunnett's T3 post-hoc test was used to account for multiple comparisons in ANOVA, and Dunn's post-hoc test was

used for non-parametric tests. Data and statistical tests are reported in supplementary tables for each figure. Symbols for P values are used in the figures as follows: *** (P <0.001), ** (P ≤0.01), * (P ≤0.05), ns (P ≥0.05).

Structures were visualized with PyMOL (Schrodinger LLC, 2015 [DOI](#)); *RRID:SCR_000305* [DOI](#) and figures were assembled with Inkscape v1.3 (GNU General Public License, v3; *RRID:SCR_014479* [DOI](#)).

Data availability

All analyzed data are included in supplementary tables to the manuscript. Electrophysiological source data are available from the corresponding author upon request.

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Additional information

Ethics declaration

Animal care and use and the procedures to obtain aortic tissue from mice were licensed and performed according to German animal protection law, the regulations of the state authorities of Mecklenburg-West Pomerania, and the institutional standards of the University of Rostock.

Abbreviations

β 1: regulatory β 1-subunit of BK channels
 BK α : Big Potassium (BK) channel pore-forming α -subunit (Slo1, K_{Ca} 1.1, MaxiK) BK α / β 1 complex of BK α channel and β 1 subunit
 Ca_v : voltage-gated Calcium channel
 hERG: human Ether-a-go-go Related Gene potassium channel
 IbTx: Iberiotoxin
 $\text{K}_{2\text{P}}$: Tandem-Pore-Domain Potassium channel
 K_{ir} : inward-rectifier Potassium channel
 K_v : voltage-gated Potassium channel
 MD: Molecular Dynamics (simulations)
 NCA: Negatively Charged Activator, e.g., BL-1249
 ROMK: Renal Outer Medullary Potassium channel
 SF: Selectivity Filter
 SUR: Sulfonylurea Receptor regulatory subunit of K_{ATP} channels
 THexA: Tetrahexylammonium ion
 TPA: Tetrapentylammonium ion

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Additional files

[Supplementary Information](#) 

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Peer reviews

Reviewer #1 (Public review):

In this manuscript, the authors aimed to identify the molecular target and mechanism by which α -Mangostin, a xanthone from *Garcinia mangostana*, produces vasorelaxation that could explain the antihypertensive effects. Building on on prior reports of vascular relaxation and ion channel modulation, the authors convincingly show that large-conductance potassium BK channels are the primary site of action. Using electrophysiological, pharmacological, and computational evidence, the authors achieved their aims and showed that BK channels are the critical molecular determinant of mangostin's vasodilatory effects, even though the vascular studies are quite preliminary in nature.

Strengths:

- (1) The broad pharmacological profiling of mangostin across potassium channel families, revealing BK channels - and the vascular BK- α / β 1 complex - as the potently activated target in a concentration-dependent manner.
- (2) Detailed gating analyses showing large negative shifts in voltage-dependence of activation and altered activation and deactivation kinetics.

(3) High-quality single-channel recordings for open probability and dwell times.

(4) Convincing activation in reconstituted BK α / β 1-Cav nanodomains mimicking physiological condition and functional proof-of-concept validation in mouse aortic rings.

Weaknesses are minor:

(1) Some mutagenesis data (e.g., partial loss at L312A) could benefit from complementary structural validation.

The author's rebuttal provides alphafold3 models for mutants. While there are interesting preliminary observations, the authors decided not to include these in the main manuscript, awaiting further structural validation. I concur.

(2) While Cav-BK nanodomains were reconstituted, direct measurement of calcium signals after mangostin application onto native smooth muscle could be valuable.

In their response, the authors acknowledge the importance of measuring Ca²⁺ sparks in smooth muscle cells to further validate their findings. However, this is not provided in the manuscript. Part of my earlier comment alludes to the possibility of α -Mangostin directly affecting Cav1.2 or ryanodine receptor activity, and therefore BK activity would go up. With the current provided evidence, these possibilities cannot be excluded and need to be acknowledged.

(3) The work has impact for ion channel physiology and pharmacology, providing a mechanistic link between a natural product and vasodilation. Datasets include electrophysiology traces, mutagenesis scans, docking analyses, and aortic tension recordings. The latter however are preliminary in nature.

The authors acknowledge that additional vascular physiology experiments would strengthen the argument they make. They are however unable to provide such evidence in the present manuscript. Therefore, I strongly suggest that the authors tune down the physiological implications of α -Mangostin that they include in the manuscript. I'd also suggest that "vasorelaxation" is removed from the manuscript title, given the preliminary nature of the findings.

<https://doi.org/10.7554/eLife.109479.2.sa3>

Reviewer #2 (Public review):

Summary:

In the present manuscript, Cordeiro et al. show that α -mangostin, a xanthone obtained from the fruit of the *Garcinia mangostana* tree, behaves as an agonist of the BK channels. The authors arrive at this conclusion by examining the effects of mangostin on macroscopic and single-channel currents elicited by BK channels formed by the α subunit and α + β 1 subunits, as well as $\alpha\beta$ 1 channels coexpressed with voltage-dependent Ca²⁺ (CaV1,2) channels. The single-channel experiments show that α -mangostin produces a robust increase in the probability of opening without affecting the single-channel conductance. The authors contend that α -mangostin activation of the BK channel is state-independent, and molecular docking and mutagenesis suggest that α -mangostin binds to a site in the internal cavity. Importantly, α -mangostin (10 μ M) alleviates noradrenaline-induced contracture. Mangostin is ineffective if the contracted muscles are pretreated with the BK toxin iberiotoxin.

In this revised version of the manuscript by Cordeiro et al., the authors have adequately answered my previous concerns. However, as I stated in my comments, without determining the probability of opening across a wide range of voltages, any conclusion about the drug's

mechanism of action can be questioned. For example, the statement in Discussion line 481: "The higher shift observed in 1 μM Ca^{2+} may reflect the steep Ca^{2+} -dependence of the closed-open equilibrium (Cui, Cox and Aldrich, 1997) and the allosteric coupling of voltage and Ca^{2+} signals (Horrigan and Aldrich, 2002; Magleby, 2003; Clay, 2017), which are effective in this concentration range, which may lead to a higher apparent activation when voltage activation is facilitated by Ca^{2+} (Sun and Horrigan, 2022)." has no support in the data and is not predicted by the allosteric model. In order to have a larger shift induced by the drug in the presence of Ca^{2+} , you need either to alter the Ca^{2+} binding or the allosteric coupling factor C.

Please note that in the manuscript, there are several problems with the English in this sentence.

Minor

In Figure 1E, BKa should read BKalpha.

<https://doi.org/10.7554/eLife.109479.2.sa2>

Reviewer #3 (Public review):

Summary:

This research shows that a-mangostin, a proposed nutraceutical, with cardiovascular protecting properties, could act through the activation of large conductance potassium permeable channels (BK). The authors provide convincing electrophysiological evidence that the compound binds to BK channels and induces a potent activation, increasing the magnitude of potassium currents. Since these channels are important modulators of the membrane potential of smooth muscle in vascular tissue, this activation leads to muscle relaxation, possibly explaining cardiovascular protecting effects.

Strengths:

The authors have satisfactorily answered my previous comments and present evidence based on several lines of experiments that a-mangostin is a potent activator of BK channels. The quality of the experiments and the analysis is high and represents an appropriate level of analysis. This research is timely and provides a basis to understand the physiological effects of natural compounds with proposed cardio protective effects.

Weaknesses:

The identification of the binding site continues to be the least developed point of the manuscript. The authors show that the binding site is probably located in the hydrophobic cavity of the pore and show that point mutations reduce the magnitude of the negative voltage shift of activation produced by a-mangostin. This binding site should be demonstrated in the future using structural techniques such as cryo-EM.

<https://doi.org/10.7554/eLife.109479.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

*In this manuscript, the authors aimed to identify the molecular target and mechanism by which α -Mangostin, a xanthone from *Garcinia mangostana*, produces vasorelaxation that could explain the antihypertensive effects. Building on prior reports of vascular relaxation and ion channel modulation, the authors convincingly show that large-conductance potassium BK channels are the primary site of action. Using electrophysiological, pharmacological, and computational evidence, the authors achieved their aims and showed that BK channels are the critical molecular determinant of mangostin's vasodilatory effects, even though the vascular studies are quite preliminary in nature.*

Strengths:

- (1) The broad pharmacological profiling of mangostin across potassium channel families, revealing BK channels - and the vascular BK- α /beta1 complex - as the potentially activated target in a concentration-dependent manner.*
- (2) Detailed gating analyses showing large negative shifts in voltage-dependence of activation and altered activation and deactivation kinetics.*
- (3) High-quality single-channel recordings for open probability and dwell times.*
- (4) Convincing activation in reconstituted BK α / β 1-Ca_v nanodomains mimicking physiological conditions and functional proof-of-concept validation in mouse aortic rings.*

We thank the reviewer for acknowledging the strength of the different aspects investigated in our study.

Weaknesses are minor:

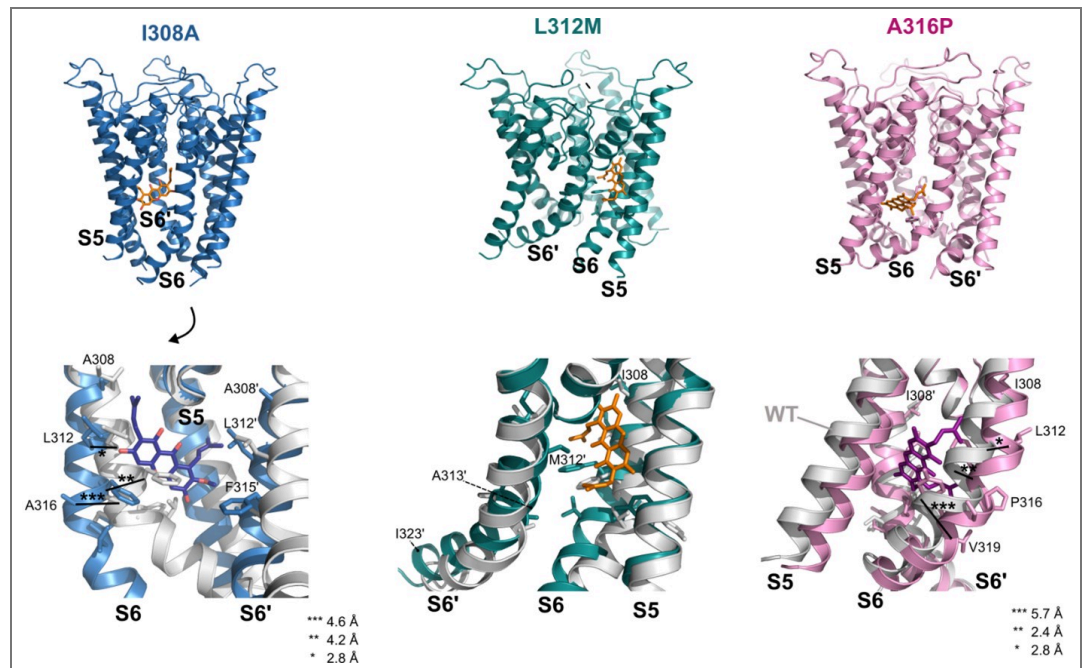
- (1) Some mutagenesis data (e.g., partial loss at L312A) could benefit from complementary structural validation.*

In the attempt to improve structural insight for the presented mutagenesis data, we have used AlphaFold3 (AF3; Abramson et al., 2024) to generate models of the I308A, L312M and A316P substitutions and repeated the docking for each (Fig. R1). According to these predictive models,

The I308A substitution considerably straightens the S6 helix starting at this residue. Hence, all residues are displaced relative to the WT: C _{α} of L312, F315, and A316 are displaced by 2.8 Å, 4.2 Å, and 4.6 Å, respectively, widening the bottom of the binding pocket. However, the prediction confidence is rated lower as in the other AF3 models for all helices (70 > pLDDT > 50). In the docking, poses in the binding pocket comparable to these observed in the WT (i.e. involving I308A, L312 and A316) and with the same molecule orientation have higher binding energies (-7.13 to -6.66 kcal mol⁻¹). Additionally, poses without contact to I308A arise that have a more vertical position, indicating that the structural change affects the binding region.

The changes induced by L312M are localized to residues 313-323, where S6 bends towards S5. Binding energies are lower especially in the best 2 poses that are also most comparable to the WT docking (-9.88 kcal mol⁻¹), but clustering overall is poor and poses are more heterogeneous. Interactions with L312M are completely abolished, while interactions with I308 (in 11/20 poses), F315 (in all poses), and A316 (in 5/20 poses) persist. Because of the rather small structural alteration induced by the substitution and the variable poses one could speculate that the reduced V_{1/2} shift is due to the observed loss in binding to L312M; however, retained interactions to the other residues would still allow α -Mangostin to activate.

A316P induces a displacement of the S6 helix compared to the WT while the other pore helices are not affected. S6 shows an enhanced outward bending around A316, which results in displacements of residues where α -Mangostin would bind, i.e., the C_a of F315 and L312M are displaced by 2.4 Å and 2.8 Å (I308 is not affected). Residues below are moved in a more rotational way, resulting in a C_a displacement of 3.1 Å for Y318 and even 5.7 Å for V319, before displacements decrease again towards the intracellular helix end. While interactions with A316P are present in 10/20 analyzed poses, the helix displacement seems to hinder I308 and L312 interactions, as the best docked α -Mangostin pose (-8.41 kcal mol⁻¹) is predicted to only contact F315 and Y318, and overall, any I308 or L312 contacts only occurred in 3/20 and 7/20 poses (wildtype: 17/20 and 20/20 poses). This may hint at a mechanism where A316P probably has a substantial allosteric share in reducing the V_{1/2} shift induced by α -Mangostin and underlines the exceptional effect of this mutation (i.e., complete loss of a V_{1/2} shift).



Author response image 1. AlphaFold3 models of BK I308A, L312M, and A316P with α -Mangostin docked to the mutant structures. The upper row shows an overview of the mutant pore helices (AF3 models) used for molecular docking. The lower row shows the binding region with the wildtype structure overlaid in gray. Only 3 helices are shown for clarity.

Although these results provide interesting tentative explanations for the effect of the mutations and conclusions from AF3 models become increasingly robust, we think that definitive statements of their mechanistic contributions would require experimental studies of mutant channels, i.e., cryo-EM or crystallography, that are beyond our means. Therefore, we have decided not to include this data in the manuscript; however, it is accessible for the interested reader within the public review. Hopefully, as cryo-EM structures have been obtained for the wildtype channel, there will be studies on mutations of this gating-relevant S6 segment in the future.

(2) While Cav-BK nanodomains were reconstituted, direct measurement of calcium signals after mangostin application onto native smooth muscle could be valuable.

We are not sure if a global elevation of cellular calcium concentration would be informative. We rather expect that the relevant local Ca²⁺ elevation would occur as sparks in the BK-Ca_v nanodomains, close to the membrane. We would anticipate a change in spark duration, as the

Ca²⁺ inward current would be stopped faster by the enhanced repolarization via α -Mangostin activated BK α / β 1 channels. This would require fast Ca²⁺ imaging acquisition speed to capture spark activity. We concur that this would be an informative experiment to investigate a more native situation. However, we would have to accomplish such methodologically challenging measurements in a separate project, which could fruitfully be combined with a more extensive characterization of aortic contraction as also suggested in the following remark (3).

(3) The work has an impact on ion channel physiology and pharmacology, providing a mechanistic link between a natural product and vasodilation. Datasets include electrophysiology traces, mutagenesis scans, docking analyses, and aortic tension recordings. The latter, however, are preliminary in nature.

We completely agree with the reviewer that there is ample room for further studies that could characterize different tissues important in blood pressure regulation (such as resistance arteries), elucidate even more physiological detail (such as modulatory effects of the endothelium), or look deeper into the pharmacology using chemically altered Mangostin derivatives. While we very much like this to happen in future projects, in this study we focused on the functional aspects of α -Mangostin in BK channel gating. We present our tension recordings as a proof-of-concept to underline the activity of α -Mangostin in native tissues, and we clearly show the importance of the BK channel by using iberiotoxin as a specific inhibitor which impressively abolished relaxation.

References:

Abramson, J. et al. (2024) "Accurate structure prediction of biomolecular interactions with AlphaFold 3," Nature, 630(8016), pp. 493–500. Available at: <https://doi.org/10.1038/s41586-024-07487-w> .

Reviewer #2 (Public review):

Summary:

*In the present manuscript, Cordeiro et al. show that α -mangostin, a xanthone obtained from the fruit of the *Garcinia mangostana* tree, behaves as an agonist of the BK channels. The authors arrive at this conclusion through the effect of mangostin on macroscopic and single-channel currents elicited by BK channels formed by the α subunit and $\alpha + \beta$ 1 subunits, as well as $\alpha\beta$ 1 channels coexpressed with voltage-dependent Ca²⁺ (CaV1,2) channels. The single-channel experiments show that α -mangostin produces a robust increase in the probability of opening without affecting the single-channel conductance. The authors contend that α -mangostin activation of the BK channel is state-independent and molecular docking and mutagenesis suggest that α -mangostin binds to a site in the internal cavity. Importantly, α -mangostin (10 μ M) alleviates the contracture promoted by noradrenaline. Mangostin is ineffective if the contracted muscles are pretreated with the BK toxin iberiotoxin.*

Strengths:

The set of results combining electrophysiological measurements, mutagenesis, and molecular docking reveals α -mangostin as a potent activator of BK channels and the putative location of the α -mangostin binding site. Moreover, experiments conducted on aortic preparations from mice suggest that α -mangostin can aid in developing drugs to treat a myriad of diverse diseases involving the BK channel.

We thank the reviewer for pointing out the significance of our study.

Weaknesses:

Major:

(1) Although the results indicate that α -mangostin is modifying the closed-open equilibrium, the conclusion that this can be due to a stabilization of the voltage sensor in its active configuration may prove to be wrong. It is more probable that, as has been demonstrated for other activators, the α -mangostin is increasing the equilibrium constant that defines the closed-open reaction (L in the Horrigan, Aldrich allosteric gating model for BK). The paper will gain much if the authors determine the probability of opening in a wide range of voltages, to determine how the drug is affecting (or not), the channel voltage dependence, the coupling between the voltage sensor and the pore, and the closed-open equilibrium (L).

We would like to take the opportunity to clarify this potential misunderstanding. In our manuscript, we have discussed three mechanistic explanations for the Mangostin activation: (1) an electrostatic effect at the selectivity filter, (2) structural and electrostatic changes of S6 that facilitate the opening of a putative lower gate, and (3) hydrophobic gating, i.e., counteracting dewetting of the pore. All possibilities would impact S6 and lower the free energy for pore opening, and we concur that therefore Mangostin most likely affects the closed-open equilibrium (L) of the BK α channel.

The sentence at the original lines 470-471, “(...) caused by an enhanced shift of the closed-open equilibrium toward the open state, such as the stabilization of the voltage sensor in an active conformation” refers to the observation that the presence of the β 1 subunit enhances this closed-open shift. The stabilization of the voltage sensor domain was mentioned as one example of how it achieves this. We recognize that this example was an unfortunate choice, as β 1 rather facilitates Ca²⁺-dependent allosteric pore opening unrelated to the discussed mechanisms of Mangostin. We have therefore removed this statement.

As to the suggestion to dissect the effect of Mangostin on C, D, and L, we agree with the reviewer that this would surely add to a full biophysical characterization. However, in our project, we strove towards including more experiments showing the physiological implications of Mangostin activation to emphasize the implication for vasodilation. We hope the reviewer understands that, with limited resources, this came at the expense of a full investigation of the different gating components, which could pose a separate project by itself.

(2) Apparently, the molecular docking was performed using the truncated structure of the human BK channel. However, it is unclear which one, since the PDB ID given in the Methods (6vg3), according to what I could find, corresponds to the unliganded, inactive PTK7 kinase domain. Be as it may, the apo and Ca²⁺ bound structures show that there is a rotation and a displacement of the S6 transmembrane domain. Therefore, the positions of the residues I308, L312, and A316 in the closed and open configurations of the BK channel are not the same. Hence, it is expected that the strength of binding will be different whether the channel is closed or open. This point needs to be discussed.

We apologize for the typing error and thank the reviewer for indicating this erroneous PDB ID. (“6vg3”). It should have read PDB ID 6v3g as in the legend to Fig. 4B. The reviewer appropriately points out that there are differences in the S6 segment addressed in our study between the two available cryo-EM structures obtained in the presence (PDB ID 6v38) and absence of Ca²⁺ (PDB ID 6v3g) (Tao and MacKinnon, 2019).

We had actually performed the docking with both structures, but chosen to show the Ca²⁺-free structure to better visualize the I308 position. α -Mangostin is found in the same S6 region in both, not obstructing the K⁺ conduction pathway. The binding energies of the favored poses are very similar; the binding energy in the best-ranking conformational cluster in the

Ca²⁺-bound structure even was slightly lower (-8.64 kcal mol⁻¹) than in the docking with the Ca²⁺-free channel (-8.58 kcal mol⁻¹; Fig. 4B), which may not be a relevant difference.

We compared the residue interactions in both dockings (Author response table 1). S317 and Y318, which did not reduce the shift in $V_{1/2}$ upon substitution, were not predicted to contact α -Mangostin in either structure. In both structures, L312 and F315 were predicted to interact in virtually all poses analyzed. In the docking to the Ca²⁺-free state, also I308 was predicted to interact in 17/20 poses, while contacts to A316 occurred in 5/20 poses. In the Ca²⁺-bound state, predicted interactions shifted from I308 (which is expected as it is buried in the protein) to A316, and the isoprenyl moiety close to I308 rotated downwards. This could indicate that α -Mangostin adopts a more horizontal position following the upward reorientation of S6 in the Ca²⁺-bound state when the channel moves from one to the other conformation (Fig. S4).

Residue	Ca ²⁺ -free structure (6v3g)	Ca ²⁺ -bound structure (6v38)
I308	17/20 poses	0/20 poses
L312	20/20	20/20
F315	20/20	19/20
A316	5/20	19/20
S317	0/20	0/20
Y318	0/20	0/20

Author response table 1. Number of interactions of S6 residues in 20 analyzed α -Mangostin poses in the molecular dockings to the Ca²⁺-free and Ca²⁺

These docking results are consistent with our functional measurements. Recent structures of the BK/ γ 1 complex showed that the VSD and Ca²⁺-bowl are stabilized in an active-like conformation that corresponds to the conformation seen in the Ca²⁺-bound state (Kallure et al., 2023; Yamanouchi et al., 2023; Redhardt, Raunser and Raisch, 2024), indicating that very likely the Ca²⁺-bound and Ca²⁺-free structures indeed represent open and closed conformations of the channel. We observed that α -Mangostin can bind to both of these states to activate the channel (Fig. 3C, D), showing the presence of a binding site in both conformations. Further, α -Mangostin induced a left-shift in $V_{1/2}$ also in higher Ca²⁺ concentration (Fig. 2D), indicating that it still binds to and activates the channel after the conformational change in S6. As we could not determine affinity for the mutants due to limited solubility, we have no information on the nature of the contribution of the substitutions, i.e., reduced binding or allosteric effect. As I308 is buried in the Ca²⁺-bound state, its contribution is likely mostly allosteric. We have also proposed dewetting as possible activation mechanism, which we expect to be less sensitive to the exact pose of a molecule (as shown for NS11021, Nordquist et al., 2024). Therefore, α -Mangostin could, e.g., change solvent accessibility of the I308 sidechain, energetically favoring the buried (open) state.

We have now included both dockings and Author response table 1 in Fig. S4, and we have added passages to the results section (starting at line 373) and discussion section (starting at lines 544, 588).

Minor:

(1) From Figure 3A, it is apparent that the increase in P_o is at the expense of the long periods (seconds) that the channel remains closed. One might suggest that α -mangostin increases the burst periods. It would be beneficial if the authors measured both closed and open dwell times to test whether α -mangostin primarily affects the burst periods.

We thank the reviewer for this valuable suggestion, which we have implemented. In our single channel measurements shown in our original Fig. 3 we have not observed burst behavior of the BK α channels. This can be explained by the fact that we measured in resting condition (100 nM free Ca $_i$ 2+) and with rather mild depolarisation (+40 mV) where Po was very low. We have therefore analyzed measurements in 5 μ M free a $_i$ 2+ where we recorded sufficient burst activity also in the basal state.

The burst analysis showed that α -Mangostin indeed prolongs bursts and shortens the interburst closures. Within bursts, both closed times and open times were increased, and we recorded a higher number of opening events per burst. We conclude that α -Mangostin acts in both the closed and the open state, where it slows open-closed transitions resulting in less flicker, and stabilizes the open state via longer open times and a higher probability for closed-open transitions.

We now show this data in Fig. 3D-F and Table S8, and have accordingly added passages to the results section (starting at line 285), the discussion (line 510), and the methods section (starting at line 746).

(2) In several places, the authors make similarities in the mode of action of other BK activators and α -mangostin; however, the work of Gessner et al. PNAS 2012 indicates that NS1619 and Cym04 interact with the S6/RCK linker, and Webb et al. demonstrated that GoSlo-SR-5-6 agonist activity is abolished when residues in the S4/S5 linker and in the S6C region are mutated. These findings indicate that binding of the agonist is not near the selectivity filter, as the authors' results suggest that α -mangostin binds.

We will gladly clarify our ideas concerning the binding sites of other activators and α -Mangostin. We first hypothesized that α -Mangostin may share characteristics and mode of action with the class of negatively charged activators (NCA) that we have described before (Schewe et al., 2019). NCA were found to occupy a common fenestration site that is located close to the selectivity filter in TREK K2P channels, and in this manuscript we have shown by THexA competition and mutagenesis experiments that α -Mangostin also binds in this fenestration region in TREK-1 channels (Fig. S3).

The existence of this common NCA binding site was also proposed for BK channels, as a docking placed the NCA NS11021 in an equivalent binding region, and, among others, NS11021 and GoSlo-SR-5-6 competed with THexA for binding in the pore (Schewe et al., 2019). These results were indeed not fully in agreement with the proposed binding site of GoSlo-SR-5-6 in Webb et al. (2015), although the most effective (double) mutants were located at S317 and I323, at the intracellular end of the cleft between neighboring S6 segments. In this manuscript, we have shown that α -Mangostin is present in the pore of BK channels by molecular docking, a THexA competition assay, and two mutations that reduced the shift in $V_{1/2}$ induced not only by α -Mangostin but also by GoSlo-SR-5-6 (Fig. 4). While the docking was rather a starting point, both functional tests argue against a binding site in the S4/5 linker/S6C region; however, allosteric mechanisms could still reduce activation also in mutants in the S4/5 linker/S6C region far from the pore binding region proposed by us in the 2019 study and the present manuscript.

To summarize, we did not mean to imply that all BK activators should bind to this site, especially if they are not part of the NCA class (as NS1619, Cym4, as well as BC5, whose different binding site enabled us to use it as a control in our THexA competition assay). However, the cleft close to gating relevant S6 residues may well pose a region especially susceptible to modulator binding (as BL-1249, GoSlo-SR-5-6, and α -Mangostin). We have moved, respectively separated, the initial GoSlo references from the reference to the pore binding site in the paragraph (lines 329, 358) to improve clarity.

(3) The sentence starting in line 452 states that there is a pronounced allosteric coupling between the voltage sensors and Ca^{2+} binding. If the authors are referring to the coupling factor E in the Horrigan-Aldrich gating model, the references cited, in particular, Sun and Horrigan, concluded that the coupling between those sensors is weak.

We are grateful for the opportunity to improve this passage. We intended to express that observed effects (in this case the shift in $V_{1/2}$) are pronounced around $1\ \mu\text{M}\ \text{Ca}^{2+}$. As the reviewer states, the coupling factor between the voltage and calcium sensors (E ; 2.4) is weak compared to the coupling of Ca^{2+} (C; 8) and voltage (D; 25) to the pore in the Horrigan-Aldrich model. However, the shape of the Ca^{2+} -dependence of $V_{1/2}$ cannot be completely described when E is neglected, with the highest difference around $1\text{--}2\ \mu\text{M}\ \text{Ca}^{2+}$ (Horrigan and Aldrich, 2002). Deletion of the gating ring underlines the allosteric sensor coupling (Clay, 2017). This together with the steep Ca^{2+} -dependence in this concentration range (meaning high P_o changes upon occupancy increase; Cui, Cox and Aldrich, 1997) explains the higher apparent activation, visible as the higher shift in $V_{1/2}$ observed at the $1\ \mu\text{M}\ \text{Ca}^{2+}$. Speaking with the model of Sun and Horrigan (2022), the suppressing “molecular logic gate” is already relieved by the presence of intermediate Ca^{2+} , and the direct “gating lever” pathway via voltage acts synergistically and achieves the observed higher $V_{1/2}$ shift upon depolarization. We have adapted the sentence and separated the citations for better understanding (lines 503–507).

References:

Clay, J.R. (2017) “Novel description of the large conductance Ca^{2+} -modulated K^+ channel current, BK, during an action potential from suprachiasmatic nucleus neurons,” *Physiological Reports*, 5(20), p. e13473. Available at: <https://doi.org/10.14814/phy2.13473> .

Cui, J., Cox, D.H. and Aldrich, R.W. (1997) “Intrinsic Voltage Dependence and Ca^{2+} Regulation of mslo Large Conductance Ca -activated K^+ Channels,” *Journal of General Physiology*, 109(5), pp. 647–673. Available at: <https://doi.org/10.1085/jgp.109.5.647> .

Horrigan, F.T. and Aldrich, R.W. (2002) “Coupling between voltage sensor activation, Ca^{2+} binding and channel opening in large conductance (BK) potassium channels,” *The Journal of General Physiology*, 120(3), pp. 267–305. Available at: <https://doi.org/10.1085/jgp.20028605> .

Kallure, G.S. et al. (2023) “High-resolution structures illuminate key principles underlying voltage and LRRC26 regulation of Slo1 channels.” *bioRxiv*, p. 2023.12.20.572542. Available at: <https://doi.org/10.1101/2023.12.20.572542> .

Nordquist, E.B., Jia, Z., Chen, J., 2024. “Small Molecule NS11021 Promotes BK Channel Activation by Increasing Inner Pore Hydration.” *J. Chem. Inf. Model.* 64, 7616–7625. <https://doi.org/10.1021/acs.jcim.4c01012> .

Redhardt, M., Raunser, S. and Raisch, T. (2024) “Cryo-EM structure of the Slo1 potassium channel with the auxiliary $\gamma 1$ subunit suggests a mechanism for depolarization-independent activation,” *FEBS Letters*, 598(8), pp. 875–888. Available at: <https://doi.org/10.1002/1873-3468.14863> .

Schewe, M. et al. (2019) “A pharmacological master key mechanism that unlocks the selectivity filter gate in K^+ channels,” *Science*, 363(6429), pp. 875–880. Available at: <https://doi.org/10.1126/science.aav0569> .

Sun, L. and Horrigan, F.T. (2022) “A gating lever and molecular logic gate that couple voltage and calcium sensor activation to opening in BK potassium channels,” *Science Advances*, 8(50), p. eabq5772. Available at: <https://doi.org/10.1126/sciadv.abq5772> .

Tao, X. and MacKinnon, R. (2019) “Molecular structures of the human Slo1 K⁺ channel in complex with β 4,” eLife 8, p. e51409. Available at: <https://doi.org/10.7554/eLife.51409> .

Webb, T.I. et al. (2015) “Molecular mechanisms underlying the effect of the novel BK channel opener GoSlo: Involvement of the S4/S5 linker and the S6 segment,” Proceedings of the National Academy of Sciences, 112(7), pp. 2064–2069. Available at: <https://doi.org/10.1073/pnas.1400555112> .

Yamanouchi, D. et al. (2023) “Dual allosteric modulation of voltage and calcium sensitivities of the Slo1-LRRC channel complex,” Molecular Cell, 83(24), pp. 4555–4569.e4. Available at: <https://doi.org/10.1016/j.molcel.2023.11.005> .

Reviewer #3 (Public review):

Summary:

This research shows that α -mangostin, a proposed nutraceutical, with cardiovascular protective properties, could act through the activation of large conductance potassium permeable channels (BK). The authors provide convincing electrophysiological evidence that the compound binds to BK channels and induces a potent activation, increasing the magnitude of potassium currents. Since these channels are important modulators of the membrane potential of smooth muscle in vascular tissue, this activation leads to muscle relaxation, possibly explaining cardiovascular protective effects.

Strengths:

The authors present evidence based on several lines of experiments that α -mangostin is a potent activator of BK channels. The quality of the experiments and the analysis is high and represents an appropriate level of analysis. This research is timely and provides a basis to understand the physiological effects of natural compounds with proposed cardio-protective effects.

We sincerely thank the reviewer for appraising the achievements of our study.

Weaknesses:

The identification of the binding site is not the strongest point of the manuscript. The authors show that the binding site is probably located in the hydrophobic cavity of the pore and show that point mutations reduce the magnitude of the negative voltage shift of activation produced by α -mangostin. However, these experiments do not demonstrate binding to these sites, and could be explained by allosteric effects on gating induced by the mutations themselves.

We are aware that our functional data are unfortunately not sufficient to clearly distinguish between effects due to affinity loss or due to allosteric mechanisms. Our attempts to generate complete dose–response curves for the mutants to determine accurate apparent IC₅₀ values were unfortunately limited by the solubility of the compound. Consequently, we have avoided making claims about affinity loss in the mutant analysis, and have instead only reported the reduction in potency, expressed as the shift in V_{1/2}. To reduce confounding effects from the mutations themselves, we selected substitutions that preserved the most wildtype-like GV-relationships, based on the extensive mutagenesis work of (Chen, Yan and Aldrich, 2014). We address this matter also in our answer to Recommendation (6) below, and we have replaced the word “binding” in the title of the manuscript. Nevertheless, we consider the proposed binding region to be well supported by the THexA competition experiments in combination with molecular docking, even though the specific mechanistic contributions of individual residues cannot yet be resolved.

Reviewer #3 (Recommendations for the authors):

(1) Natural xanthenes as α -Mangostin induce vasorelaxation via binding to key gating residues in the S6 domain of BK channels.

(2) If α -Mangostin occupies a similar binding site to quaternary ammoniums, what is the explanation for not observing a reduction in the single-channel current (fast blocking effect)? The α -Mangostin site proposed here is in a region of the channel that should occlude ion permeation. The authors should discuss possible explanations for this apparently contradictory observation.

As the reviewer states, we indeed have not observed a reduced single channel amplitude in any measurement. The THexA competition assay showed that α -Mangostin is present in the pore cavity and interferes with THexA access to its binding site. However, we do not think that their binding sites are similar, as QA ions bind directly below the filter entrance to block permeation, while our studies suggest that α -Mangostin binds in the upper portion of the cleft between S6 helices. In this position, it would clearly overlap with the QA binding site and hinder access, but not block permeation. We would therefore not expect to see an amplitude reduction by intermittent α -Mangostin block. Consistently, all binding poses in our dockings were close to the cavity wall, without interfering with the central ion conduction pathway. To better illustrate this, we have added updated intracellular views of the dockings in the Ca^{2+} -free and Ca^{2+} -bound state (which we have also now included as suggested by another reviewer) to the supplementary information (Fig. S4A).

(3) In Figure 2D, it is difficult to appreciate the differences between the symbols representing the G-V relationships of BK channels at different intracellular Ca concentrations, before and after activation with 10 μM α -Mangostin. A clearer distinction between the curves would help to interpret the data more easily.

We thank the reviewer for the suggestion to improve figure accessibility. We have changed the line appearance for better discrimination of the overlying portions.

(4) Both THexA and TPA block BK channels through voltage and state-dependent mechanisms. Therefore, their apparent affinity could change if α -Mangostin simply increases open probability or alters dwell times rather than physically blocking access to the binding site.

The reviewer addresses valid limitations that can affect the meaningfulness of competition experiments under certain conditions. However, we think that this does not apply to our results:

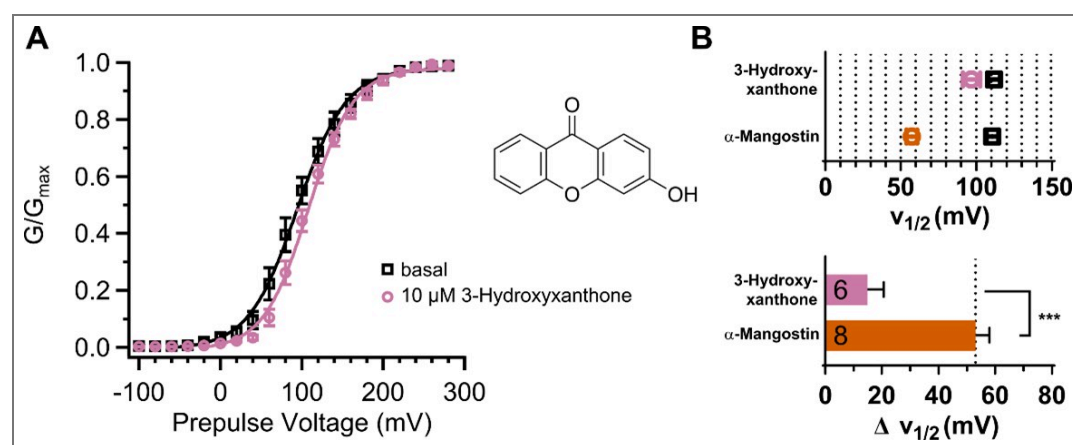
Previous studies have shown that the voltage dependence of quaternary ammonium blockers up to C_{10} is rather weak in BK channels, and only a slight increase in block is present in the voltage range +30 mV to +100 mV (Li and Aldrich, 2004; Thompson and Begenisich, 2012). Hence, THexA voltage dependence has already reached a plateau in the competition assay (at +40 mV), and its voltage dependence would have little effect on our results.

Controversy exists about the nature of the state dependence of different quaternary ammonium blockers, but TBA is often recognized as an open channel blocker of BK channels, which probably also applies to THexA (Wilkins and Aldrich, 2006; Tang, Zeng and Lingle, 2009; Thompson and Begenisich, 2012; Posson, McCoy and Nimigeon, 2013). Assuming such an open-channel block, apparent IC_{50} values would be inversely proportional to P_o . The THexA IC_{50} was about 80 nM in the basal state, when P_o is very low (0.024 at +40 mV as derived from the GV-relationship); an increase of open dwell times, respectively P_o , in the presence of α -Mangostin to, e.g., 0.3 would therefore lead to a ≈ 10 -fold decrease in apparent IC_{50} . However, the apparent THexA IC_{50} strongly increased rather than decreased (more than

20-fold to around 1.6 μM). This cannot arise from P_o change and must reflect the altered access of THexA to its binding site caused by α -Mangostin. Assuming a pure closed channel block where apparent IC_{50} would correlate with the closed times, an increase of about 1.4-fold is expected. However, we recorded a much stronger 20-fold increase. Therefore, we are convinced that we have conclusively shown that α -Mangostin is present in the BK pore irrespective of the state dependence of THexA block.

(5) The pH dependence of the $V_{1/2}$ shift supports the idea that α -Mangostin becomes more negatively charged at higher pH (enhancing its effect.) However, although the data are consistent with this interpretation, additional controls such as using a non-ionizable analog or assessing solubility changes with pH would be needed to confirm that the shift is caused specifically by ionization of α -Mangostin and not by indirect pH effects on channel gating.

We agree with the reviewer that the pH experiment by itself is not sufficient to clearly tie the existence of a charge to a possible activation mechanism. We still think that this is an interesting observation and should be made known, as we have investigated the mechanism of negatively charged activators in different K^+ channel families before (Schewe et al., 2019). Unfortunately, we do not have access to uncharged derivatives mimicking the 3D conformation. From the commercially available substances, the bare xanthone backbone is completely insoluble in water. We have therefore tested the derivative 3-hydroxyxanthone as example with a minimal number of hydroxyl substituents (Author response image 2, Author response table 2). The 3-hydroxyxanthone indeed shows reduced activation compared to α -Mangostin. The shift in $V_{1/2}$ induced by 10 μM 3-hydroxyxanthone was only 14.99 ± 5.67 mV (≈ 50 mV for α -Mangostin). This supports that the presence of several (potentially) charged substituents is important for the activation mechanism. However, we have no knowledge about the efficacy of the compound or the local pK_a of the different hydroxyl groups. As the reviewer stated, systematic chemical modifications would be necessary to elucidate the importance of the charged substituent number and positions, which is not within our capabilities.



Author response image 2. Activation of BK α by 3-hydroxyxanthone. (A) GV-relationship before and after application of 10 μM 3-hydroxyxanthone. (B) $V_{1/2}$ before and after application of 10 μM 3-hydroxyxanthone compared to α -Mangostin and the resulting difference in $V_{1/2}$ ($\Delta V_{1/2}$). Measurements were conducted as described in the main manuscript with 100 nM free Ca_i^{2+} .

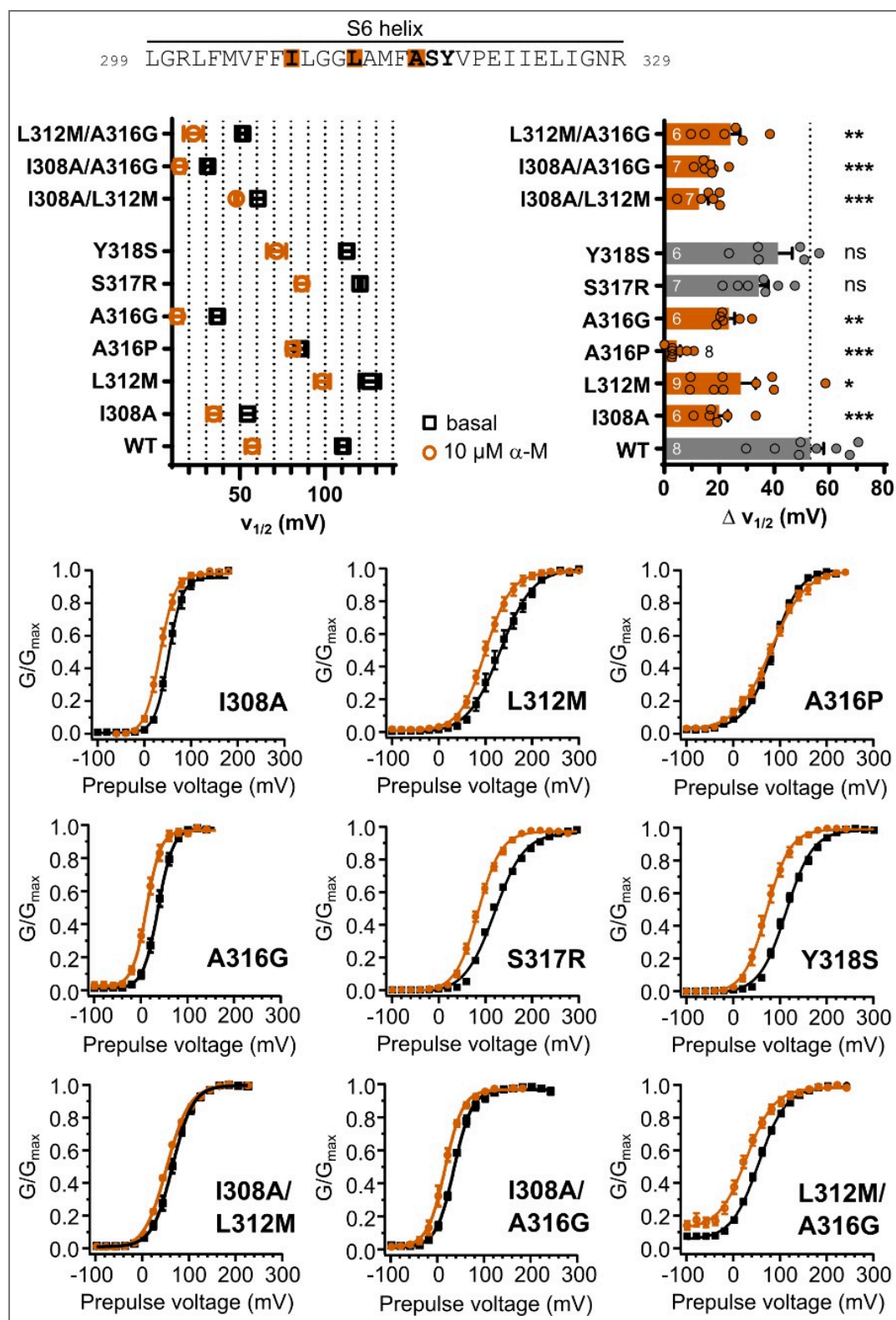
	$V_{1/2}$ (mV) basal	$V_{1/2}$ (mV) activated	$\Delta V_{1/2}$ (mV)	P-value
α -Mangostin	110.49 ± 2.69	57.37 ± 3.60	53.08 ± 4.9	<0.001
3-hydroxyxanthone	111.82 ± 3.93	96.83 ± 1.01	14.99 ± 5.67	

Author response table 2. Comparison of the $V_{1/2} \pm \text{SEM}$ and $\Delta V_{1/2} \pm \text{SEM}$ before and after activation by 10 μM α -Mangostin or 10 μM 3-hydroxyxanthone in BKa channels. Unpaired t-test, two-tailed P values ($\alpha=0.05$)

(6) The reduced $V_{1/2}$ shifts observed in the I308A, L312M, and A316PP mutants may result from intrinsic gating alterations rather than a true loss of α -Mangostin binding. The GoSlo-SR-5-6 control is informative, but the persistence of activation in A316P does not fully resolve this. A more convincing test would be employing double or triple mutants.

As stated above, we acknowledge that our functional data do not allow us to definitively separate effects arising from a true loss of binding affinity from those due to potential allosteric effects. We tried to minimize intrinsic gating alteration brought by substitutions by not conducting a pure alanine or cysteine scanning mutagenesis. Instead, substitutions were chosen to be closest to the wildtype GV-relationship in (Chen, Yan and Aldrich, 2014) where possible. While L312M was virtually identical to the wildtype, A316P showed a change in slope in high Ca^{2+} concentrations, which could indicate a changed voltage sensitivity. Additionally, A316P completely abolished α -mangostin activation. We therefore also used A316G to ensure that the channel is functional and retains voltage sensitivity, even if its $V_{1/2}$ was shifted stronger. As we have conducted paired measurements and assessed the $V_{1/2}$ before and after activation, we are confident that we can attribute a reduced shift to the reduced action of α -mangostin.

Following the reviewer's suggestion, we have generated and measured the double mutants I308A/L312M, I308A/A316G, and L312M/A316G (the triple mutant I308A/L312M/A316G did not produce measurable currents). The mutants I308A/L312M and I308A/A316G showed a moderate energy-additive effect and reduced the shift in $V_{1/2}$ by further ≈ 7 mV compared to the single mutation with the stronger shift. The combination L312M/A316G, however, did not further reduce the shift seen in the single mutations and did not even produce the shift induced by A316G alone.



Author response image 3. Double Mutants I308A/L312M, I308A/A316G and L312M/A316G compared to the single mutations in the main manuscript. The $V_{1/2}$ before and after activation with 10 μ M α -Mangostin, the resulting shift in $V_{1/2}$, and the GV-relationships are shown ($n=6-7$), measurements were made as in Fig. 4.

	$V_{1/2}$ (mV)	$V_{1/2}$ (mV) in 10 μ M α -Mangostin	$\Delta V_{1/2}$ (mV)
I308A	54.45 $\pm$ 3.86	34.48 $\pm$ 3.81	19.97 $\pm$ 3.12
L312M	126.29 $\pm$ 6.19	98.39 $\pm$ 4.46	27.89 $\pm$ 5.42
A316G	36.7 $\pm$ 2.96	13.13 $\pm$ 3.11	23.57 $\pm$ 2.05
I308A/L312M	60.57 $\pm$ 3.77	47.88 $\pm$ 0.77	12.7 $\pm$ 3.37
I308A/A316G	30.98 $\pm$ 1.94	14.47 $\pm$ 2.83	16.50 $\pm$ 1.48
L312M/A316G	51.55 $\pm$ 1.99	22.69 $\pm$ 5.81	28.86 $\pm$ 6.65

Author response table 3. Summary of the $V_{1/2}$ before and after Mangostin activation and the resulting shifts in $V_{1/2}$ for the double mutants compared to the single mutants shown in the main manuscript.

Following a suggestion by another reviewer, we have generated AlphaFold3 (AF3) models for I308A, L312M and A316P and repeated the Mangostin docking. We learned that the mutations are all predicted to substantially impact the structure of the S6 helix, therefore altering the binding region, and A316P especially impacted the nature of residue interactions. This could be an explanation why the double mutants do not show a clear and consistent additive effect.

Unfortunately, this outcome is not conclusive and the double mutants do not reveal further information compared to the single mutants. We have therefore decided not to include these measurements in the manuscript.

As we do not know if our answers will be sent to all reviewers, we repeat the relevant part about the AF3 models here:

(...) According to these predictive models,

The I308A substitution considerably straightens the S6 helix starting at this residue. Hence, all residues are displaced relative to the WT: C _{α} of L312, F315, and A316 are displaced by 2.8 Å, 4.2 Å, and 4.6 Å, respectively, widening the bottom of the binding pocket. However, the prediction confidence is rated lower as in the other AF3 models for all helices (70 > pLDDT > 50). In the docking, poses in the binding pocket comparable to these observed in the WT (i.e. involving I308A, L312 and A316) and with the same molecule orientation have higher binding energies (-7.13 to -6.66 kcal mol⁻¹). Additionally, poses without contact to I308A arise that have a more vertical position, indicating that the structural change affects the binding region.

The changes induced by L312M are localized to residues 313-323, where S6 bends towards S5. Binding energies are lower especially in the best 2 poses that are also most comparable to the WT docking (-9.88 kcal mol⁻¹), but clustering overall is poor and poses are more heterogeneous. Interactions with L312M are completely abolished, while interactions with I308 (in 11/20 poses), F315 (in all poses), and A316 (in 5/20 poses) persist. Because of the rather small structural alteration induced by the substitution and the variable poses one could speculate that the reduced $V_{1/2}$ shift is due to the observed loss in binding to L312M; however, retained interactions to the other residues would still allow α -Mangostin to activate.

A316P induces a displacement of the S6 helix compared to the WT while the other pore helices are not affected. S6 shows an enhanced outward bending around A316, which results in displacements of residues where α -Mangostin would bind, i.e., the C _{α} of F315 and L312M are displaced by 2.4 Å and 2.8 Å (I308 is not affected). Residues below are moved in a more rotational way, resulting in a C _{α} displacement of 3.1 Å for Y318 and even 5.7 Å for V319, before displacements decrease again towards the intracellular helix end. While interactions with A316P are present in 10/20 analyzed poses, the helix displacement seems to hinder I308 and L312 interactions, as the best docked α -Mangostin pose (-8.41 kcal mol⁻¹) is predicted to only contact F315 and Y318, and overall, any I308 or L312 contacts only occurred in 3/20 and 7/20 poses (wildtype: 17/20 and 20/20 poses). This may hint at a mechanism where A316P probably has a substantial allosteric share in reducing the $V_{1/2}$ shift induced by α -Mangostin and underlines the exceptional effect of this mutation (i.e., complete loss of a $V_{1/2}$ shift). (...)

(7) The subtraction approach used to isolate BK currents (difference before and after α -Mangostin) assumes that the compound affects only BK channels. However, α -Mangostin could also modulate Ca_v currents directly, as reported for other polyphenolic compounds. No vehicle (DMSO) control is shown.

We agree with the reviewer that α -Mangostin could also modulate Ca_v currents; however, this would not interfere with the conclusions drawn from this nanodomain experiment. We intended to show the overall current modulation by α -Mangostin in the voltage range relevant for Ca_v -BK coupling, as this would be the determinant for the membrane potential mediating the vasoactive effect. In native tissue, BK and Ca_v channels (among others) would likewise contribute to the net membrane conductance, with BK channels being a major contributor when activated. In fact, a concomitant inhibition of Ca_v channels could act synergistically in favor of vasodilation. This could therefore be a subject for the further investigation of potential α -Mangostin targets. However, the fact that iberiotoxin prevented relaxation in aortic preparations conclusively showed that BK channels are the major player in native tissue.

We have reformulated some sentences to prevent misunderstandings that we refer to isolated BK currents instead of α -Mangostin activated currents.

DMSO controls were conducted and did not impact BK or $\text{Ca}_v1.2$ currents or the aortic tissue contraction. We have added representative measurements as Fig. S6 and stated the DMSO concentration in the Methods section (line 655).

(8) Most kinetic fits were obtained at strong depolarizations (around +100 mV), which limits how well these results can be extrapolated to physiological voltages. Although the BK- Ca_v experiments show facilitation between -50 and +50 mV, providing plots for activation and deactivation in that range would strengthen the physiological relevance.

We thank the reviewer for this valuable suggestion. We now additionally show that the impact of α -Mangostin on activation is high at lower depolarisation, indeed underlining its physiological relevance. To address the activation time course in a more physiological voltage range, we have used our measurements of BK α channels in 10 μM Ca_i^{2+} (where the $V_{1/2}$ shift induced by α -Mangostin is equal to 100 nM Ca_i^{2+} ; Fig. 2D). The outward currents already present in the lower voltage range under these conditions allowed us to fit a monoexponential function to the traces of 0 mV to 100 mV prepulses. The τ of activation decreased from 29.6 ± 3.1 ms at 0 mV to 2.4 ± 2 ms at +100 mV. After α -Mangostin activation, the time course was accelerated, with a τ of activation of 9.5 ± 4.7 ms at 0 mV to 2 ± 0.6 ms at +100 mV. This faster activation was particularly effective in the lower voltage range far from high Po , e.g., α -Mangostin caused a decrease of more than half of the τ of activation at +20 mV (from 12.2 ± 0.6 ms to 4.98 ± 1.6 ms).

Our data consists of families of different prepulse voltages and a fixed repolarisation step (to -50 mV for 100 nM free Ca_i^{2+} , and to -100 mV for 10 μM free Ca_i^{2+}). Thus, we are not able to add plots for the voltage-dependence of deactivation in the same way as for activation. However, we can present the deactivation time constants of lower prepulse voltage steps that produce outward currents in symmetrical ion conditions with 10 μM free Ca_i^{2+} . For -20 mV and +20 mV prepulse voltages, which better reflect physiological depolarisation, the deactivation time constant shows a 3-to 5-fold increase after α -Mangostin activation.

We now show the plot for the voltage dependence of activation in Fig. S2A and a bar graph for activation/ deactivation time constants at +20 mV as Fig. S2B; data are summarized in Table S5. We hope this adds to illustrating the effect of α -Mangostin under physiological conditions.

(9) Minor: In several parts of the paper, induced shifts to negative voltages are referred to "leftward shifts". It would be useful to be consistent and employ a more specific reference to negative or positive directions.

We thank the reviewer for the careful reading and have harmonized the terminology.

References

- Chen, X., Yan, J. and Aldrich, R.W. (2014) “BK channel opening involves side-chain reorientation of multiple deep-pore residues,” *Proceedings of the National Academy of Sciences*, 111(1), pp. E79–E88. Available at: <https://doi.org/10.1073/pnas.1321697111> .
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